Literature and related work

Verified sources for the analytical part (IACS A-3, A-4): comparable theses, vision-language models versus OCR pipelines, OCR engines, LLM-based extraction, serving techniques

Project: Latency and Accuracy Optimization for OCR to LLM Document Processing Pipelines · 20 October 2026

150	8	Read first
sources, each opened and summarised	topics, matching the analytical part	cite only what you have read yourself

Each source was opened and its title, authors, venue and link checked. The summaries say what the source reports, so you can decide what to read; they are not text for the thesis. Before citing a number, check it in the source itself.

Compiled 2026-10-20. Every entry below was opened (WebFetch of the DOI/venue page, arXiv abstract/HTML, PDF, official documentation or repository record) and the quoted numbers were read on that page. Where only part of a source could be read, the entry says so. Sources already in the student's reading list (Smith 2007, LayoutLM, LayoutLMv3, DocLLM, LMDX, Donut, Pix2Struct, Qwen2-VL paper, T5, Flan-T5, Sentence-BERT, MiniLM, LLM.int8, QLoRA, GPTQ, Orca, vLLM, CORD, FUNSD, SROIE, cloud vendor OCR docs) are deliberately not repeated.

Legend. ★ = core source (read first; about 50 sources in total). Tags: [A-3/A-4 analysis], [VLM vs OCR justification], [OCR tools], [Caching], [Compression], [Async/batching], [Evaluation metrics], [Related thesis]. "arXiv preprint" = not peer-reviewed as far as could be checked. Numbers in quotation marks are verbatim; numbers without quotation marks are taken from the named table/section. Lines marked "own arithmetic" were computed by the compiler, not by the source.

Strongest arguments for keeping OCR, with sources

- **OCR text improves even large VLMs, and at equal size a text LLM on OCR text beats a VLM on key-information extraction.** GPT-4 Vision rises from 84.6 to 87.4 ANLS on DocVQA and 67.3 to 71.9 on InfographicVQA when OCR text is added to the image (Borchmann 2024, Table 1). Zero-shot SROIE: text-only Llama2-7B-chat on OCR text 68.97 vs Qwen-VL-7B 58.59 vs LLaVA-1.5-7B 3.83 (LayoutLLM, CVPR 2024, Table 1: "Generally, the existing LLMs are better than MLLMs for zero-shot document VQA and VIE"). GPT-4-turbo on OCR text 0.890 vs GPT-4-vision on the image 0.834 ANLS* on SROIE (Peer et al. 2024, Table 4). With a limited visual-token budget, adding OCR lifts InternVL2 DocVQA from 56.0% to 86.6% (DocVLM, CVPR 2025).
- **Document-capable VLMs do not fit a 4 GB GPU; the thesis model does.** Measured GPU memory: Qwen2-VL-2B 13.7 GB, InternVL2-2B 10.5 GB, SmolVLM-2.2B 4.9 GB (SmolVLM 2025, Table 1 and Sec. 4.3); PaliGemma-3B at 448 px 6.72 GB (SmolVLM blog); DeepSeek-VL2 3B "single GPU with 10 GB"; MiniCPM-V 8B "16~17G" in fp16 and "around 5G" at 4 bit; LLaVA-1.5-13B "41.6G of memory usage" (Efficient-MLLM survey). By the Hugging Face rule ("4 * X GB of VRAM in float32"), Flan-T5-base (0.248B) needs about 1 GB in fp32 (own arithmetic). Caveat: sub-1B VLMs do fit (SmolVLM-256M 0.8 GB, SmolDocling 0.489 GB, Florence-2-B 232M, GOT-OCR2.0 580M claimed to run in "4G memory"), but they are OCR/page-conversion models or much weaker at document QA (SmolVLM-256M DocVQA 58.3%) – they replace Tesseract, not the extraction step.
- **A page image costs hundreds to thousands of tokens, and document accuracy needs the expensive high resolutions.** Qwen2-VL: "visual tokens per image ... 4-16384" (model card); PaliGemma DocVQA 43.7 → 78.0 → 84.8 as tokens grow 256 → 1024 → 4096 (Beyer et al. 2024, Table 1); OCR-free document models use ~768–5,118 visual tokens per page with first-token latency up to 3.73 s (DocOwl2, ACL 2025, Tables 2–3); image tokens are on average 64% of LLaVA's input tokens (FastV, ECCV 2024).
- **VLMs are still weaker than specialised OCR on hard text and hallucinate values, without a verifiable text location.** On handwriting "LMMs exhibit a 51.9% lower performance compared to the supervised

state-of-the-art" (OCRBench, Sec. 3.2); best LMM KIE F1 is 71.76 (CC-OCR, ICCV 2025, Table 5); on degraded receipts GPT-4o scores 30.21 and VLMs report "values not present in the image" (KIE-HVQA, NeurIPS 2025); GPT-4V "tends to generate new keys" (Shi et al. 2023); GPT-4o scores 0 on text spotting, i.e. cannot localise what it read (OCRBench v2); pipeline MinerU 0.15 vs GPT-4o 0.233 edit distance on English pages (OmniDocBench, CVPR 2025). Docling's authors: VLM converters "are prone to hallucinations" and "demand substantial computational resources" (AAAI-25 workshop).

- **OCR-filter + small model is cheaper and faster than sending whole documents to a large VLM.** PaddleOCR page filter + compact VLM vs Qwen2.5-VL-72B: "8.8 times higher field level accuracy", 0.7% of the GPU cost, 92.6% lower latency (Jin et al. 2025). Cost per million pages: GPT-4o API $12,480 vs self-hosted 7B VLM $176 vs MinerU pipeline $596 (olmOCR 2025, Table 6). Cloud MLLMs take ~1.4–7.0 s per page (Shen et al. 2026, Table 2).
- **Task-specific small models are far cheaper in energy and usually better for IE than general generative models.** Multi-purpose generative models are "orders of magnitude more expensive than task-specific systems" (Luccioni et al., FAccT 2024; Flan-T5-base 0.026 kWh per 1,000 inferences). LLMs "consistently exhibit inferior performance, higher latency, and increased budget requirements" than fine-tuned small models on 9 IE datasets (Ma et al., Findings EMNLP 2023).
- **Control, privacy and reproducibility favour a local pipeline with fixed weights.** VLM tools are "typically offered as SaaS" (Docling 2025); regulated domains "frequently prohibit the use of general-purpose external services" (OnPrem.LLM 2025); hosted LLMs set to be deterministic still show "accuracy variations up to 15% across naturally occurring runs" (Atil et al. 2024), and the same API model changed from 84% to 51% on a task within three months (Chen et al., HDSR 2024).

Counter-evidence the thesis must address (honest framing): "OCR may not be necessary for powerful MLLMs" – image-only Gemini 1.5 Pro gives the best F1 (76.8) on ~1,000 real business documents (Shen et al. 2026); VLMs beat pipelines on handwritten notes (OmniDocBench); OCR noise alone costs at least 14% (5 F1) in RAG (OHRBench, ICCV 2025) and Tesseract text lowers a 7B LLM's Value F1 on SROIE from 0.8930 (gold) to 0.6742 (Anvari 2026); tiny VLMs such as SmolDocling (0.489 GB VRAM) now exist. The defensible claim is therefore conditional: *under a 4 GB VRAM / local / auditable-output constraint* and for *fixed fields on mostly printed invoices*.

1. Comparable theses

Verification note: DiVA (diva-portal.org) record pages could not be loaded from the compilation environment, so Swedish theses were opened in Uppsök, the national index of Swedish student theses (KB/Libris), which harvests DiVA; the links given are the official urn.kb.se persistent identifiers that resolve to the DiVA record. Uppsök level codes: H2 = two-year master's, M2 = bachelor's. Aalto records were read through Aalto's repository API (the HTML pages return 403 to automated fetches). All other records/PDFs were opened directly.

1.1 Invoice and receipt information extraction

- ★ **Kesete, A., Gustafsson, I. (2026). AI and Non-AI Approaches to Invoice Information Extraction: A Comparison in Local Deployment Contexts.** Master's thesis (Uppsök level H2), Jönköping University. http://urn.kb.se/resolve?urn=urn:nbn:se:hj:diva-72401 — [Related thesis] [Compression] [Evaluation metrics] [A-3/A-4 analysis] — Template-based Invoice2Data vs a 133M-parameter LayoutLMv3 on a consumer CPU, on IDSEM and 1,000 generated invoices. On IDSEM, templates reach 100% Exact Match while the AI model "achieves a mean of only 18.6% Exact Match, with a standard deviation of 25.0% across folds"; on the generated set LayoutLMv3 reaches 89.1%. Dynamic INT8 quantization cut model size by about 51% and latency by about 3x with <0.4 pp loss of Exact Match. (OpenAIRE labels it "Bachelor thesis"; Uppsök lists level H2.)
- ★ **Leon, D. (2021). Extracting Information From PDF Invoices Using Deep Learning.** Master's thesis (H2), KTH Royal Institute of Technology. http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-304475 — [Related thesis] [Evaluation metrics] — Extracts 4 fields from 899 PDF invoices with BERT, Faster R-CNN and a rule-based baseline; overall F1 0.911 (BERT), 0.830 (rules), 0.815 (Faster R-CNN).
- **Tan, T. (2023). Information Extraction from Invoices using Graph Neural Networks.** Master's thesis (H2), KTH Royal Institute of Technology. http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-336620 — [Related

thesis] — Invoices as graphs, GNN word-role prediction plus rule-based post-processing; best variant "90.93% correct rate".

- **MohamadAnas, H. (2026). HELIX: Hybrid Evidence-Linked Invoice Header Extraction: A Validation-Aware System for Accounting Workflows.** Master's thesis (H2), Linnaeus University. http://urn.kb.se/resolve?urn=urn:nbn:se:lnu:diva-147627 — [Related thesis] [Evaluation metrics] — Text recovery/OCR + field-specific parsers + optional local LLM + grounding checks and validation rules; on 110 labelled invoices and 14 header fields: positive-field semantic F1 95.4%, schema-level field accuracy 94.9%; invoice date and number performed especially well.
- ★ **Pereira, M. G. F. M. (2024). From regular expressions to language models: A text-based approach for invoice information extraction.** Master's thesis, Iscte – University Institute of Lisbon. http://hdl.handle.net/10071/33967 — [Related thesis] — Text-only extraction on a company invoice set: regular expressions 75% accuracy for VAT numbers; Mistral-7B-Instruct-v0.3 76% for date and 71% for total – directly comparable fields to the thesis.
- **Correia, J. E. da C. (2024). Extração automática de informação de faturas digitalizadas.** Master's thesis, Iscte – University Institute of Lisbon. http://hdl.handle.net/10071/33869 — [Related thesis] — Tuned LayoutLMv3 vs an existing LayoutLMv2-based system on scanned invoices; LayoutLMv3 F1: invoice number 90%, invoice date 91%, supplier name 91%; the older system was better on detailed fields (quantity F1 94%).
- ★ **Silva, R. M. V. da (2025). Fine-tuning a Multimodal Machine Learning Model for Key Information Extraction from Invoices and Receipts.** Master's thesis, NOVA IMS, Universidade Nova de Lisboa. https://hdl.handle.net/10362/190490 — [Related thesis] [OCR tools] [Evaluation metrics] — Tesseract OCR + LayoutLMv3 fine-tuned on 813 Portuguese invoices/receipts vs Google Document AI. Table 4.1: overall F1 0.30 (LayoutLMv3) vs 0.78 (Document AI); date 0.40 vs 0.96; total 0.78 vs 0.89; low recall was the main weakness.
- **Leander, A. (2023). Using Transformers to Extract Date Expressions From Receipts.** Master's thesis, Lund University (LU-CS-EX 2023-33). https://lup.lub.lu.se/student-papers/record/9138650 — [Related thesis] [Evaluation metrics] — Regex system vs two pre-trained transformers trained on regex-labelled and synthetic data: test accuracy 94.62% (transformers) vs 96.79% (regex); the ML system sometimes coped with OCR misses and unseen date formats.
- **Norman, P. (2024). Key Information Extraction From Swedish Receipts Using Multimodal Machine Learning Models.** Master's thesis (H2), KTH Royal Institute of Technology. http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-353035 — [Related thesis] [OCR tools] — LayoutLMv3, PICK and ViBERTgrid on 299 Swedish and 1,000 foreign receipts with corrected and uncorrected OCR; trained only on foreign data, micro F1 71.49 (PICK), 66.95 (ViBERTgrid), 52.60 (LayoutLMv3).

1.2 LLM/VLM vs OCR pipelines

- ★ **Bardvall, M., Hassle, I. (2024). Automating Invoice Recognition: A Comparative Study of Large Language Models and OCR/ML Technologies.** Bachelor's thesis (M2), KTH Royal Institute of Technology. http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-351177 — [VLM vs OCR justification] [Related thesis] — GPT vision model vs Azure AI Document Intelligence (OCR + ML) on invoices: Azure "generally perform better in accurately handling invoices"; GPT's performance "commendable" but "Azure leads in task-specific accuracy". No numbers in the abstract.
- **Autonen, A. (2026). Assessing the role of LLM-based OCR in enhancing invoice data capture accuracy in accounts payable.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/145600 — [VLM vs OCR justification] [Related thesis] — Case study (27 invoices, Dooap Oy): LLM-based OCR had higher overall header-field accuracy than traditional OCR, but not for every field; it struggled with long invoices, ambiguous references and non-standard formats.
- ★ **Lähdekorpi, I. (2025). Fine-tuning transformer-based models to extract transactions from bank statements.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/140175 — [VLM vs OCR justification] [Evaluation metrics] — Text LLMs fine-tuned on OCR text vs an image-based multimodal model on 1,759 statements from 6 banks: text models in-distribution F1 > 0.7 (exact match < 0.3); the multimodal model did worse (exact match 0.05, F1 0.48 in-distribution, near zero out-of-distribution).

- **Paulsson, C., Valestrand Köster, A., Hermansson Aspenbäck, D. (2026). Evaluation of Transactional Document Extraction Methods.** Bachelor's thesis (M2), University of Skövde. http://urn.kb.se/resolve?urn=urn:nbn:se:his:diva-26561 — [VLM vs OCR justification] [Related thesis] — IDP systems vs general multimodal LLMs vs local LLM pipelines on order confirmations, invoices and receipts: IDP most accurate on known layouts, multimodal LLMs most flexible, local LLM pipelines weakest. No numbers in the abstract.
- **Daqqah, B. H. (2024). Leveraging Large Language Models (LLMs) for Automated Extraction and Processing of Complex Ordering Forms.** M.Eng. thesis, Massachusetts Institute of Technology. https://hdl.handle.net/1721.1/156567 — [VLM vs OCR justification] [Related thesis] — Page classification → OCR and table extraction → LLM inference → code combination; GPT-4V one-shot outperformed a fine-tuned GPT-3.5 for key-value extraction. No numbers in the abstract.
- **Johansson, M., Waginder, S. (2026). Vision Language Model Based Systems for Optical Character Recognition of Historical Swedish Newspaper Material.** Master's thesis, Chalmers University of Technology. https://hdl.handle.net/20.500.12380/311006 — [VLM vs OCR justification] [OCR tools] — Counterpoint: a fine-tuned Qwen3-VL-8B-Instruct + repetition trimmer beat ABBYY FineReader by 68.5% in CER (1.930% CER), but only after domain fine-tuning and with repetition handling.
- **Gomes Ziegler, G. (2024). Automating Information Extraction from Financial Reports Using LLMs.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/131752 — [VLM vs OCR justification] — Text-only vs image-only vs multimodal LLM extraction of ESG indicators: multimodal best (perfect match above 85% for key indicators); image-only suffered hallucinations.

1.3 OCR tools

- ★ **Erlandsson, Z. (2019). Suitability of OCR Engines in Information Extraction Systems: a Comparative Evaluation.** Master's thesis (H2), KTH Royal Institute of Technology. http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-255021 — [OCR tools] [Related thesis] — Tesseract vs Google Cloud Vision as the OCR stage of an extraction system for financial PDFs (115 documents): Tesseract gave statistically significantly higher end-to-end extraction accuracy, attributed to its more predictable output structure.
- **Hammarström, E. (2025). Investigation of OCR Model Performance and Post-OCR Correction Strategies: A Comparative Analysis.** Master's thesis (H2), Uppsala University. http://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-565512 — [OCR tools] [VLM vs OCR justification] [Evaluation metrics] — Tesseract, Kraken, DocTR and the Idefics VLM on 43,823 newspaper paragraphs: Tesseract 5.5.0 best overall (6.7% CER, 22.4% WER), ahead of Kraken (7.6% / 26.5%); after post-OCR correction DocTR had the lowest CER.
- **Nordin Hällgren, S. (2021). Reading Key Figures from Annual Reports.** Master's thesis, Chalmers University of Technology. https://hdl.handle.net/20.500.12380/302433 — [OCR tools] [Related thesis] — Tesseract 4.1.1 on 280 scanned annual reports + TF-IDF Random Forest page classifier (99.6% test accuracy) + approximate string matching: 89.6% extraction accuracy; a denoising filter raised accuracy from 73.8% to 83.8% on hard documents.
- **Billah, M. M. (2025). A Robust Framework for Data Extraction from Unstructured Sources Using Large Language Models.** Master's thesis (H2), Mälardalen University. http://urn.kb.se/resolve?urn=urn:nbn:se:mdh:diva-71838 — [OCR tools] [Related thesis] — Industrial PDFs incl. invoices: Tesseract and pdfplumber most promising for layout-preserving text; few-shot prompting beat zero-shot and LoRA fine-tuning; Mistral 7B reached 87.11% data retrieval.
- **Silfving, M., Liu, D. (2025). Evaluation of OCR Engines for Logistics Documents: Accuracy, Preprocessing and Cloud Integration.** Bachelor's thesis (M2), KTH Royal Institute of Technology. http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-373827 — [OCR tools] [Evaluation metrics] — CER of Tesseract, Google Cloud Vision and Amazon Textract on 20 labelled images with/without preprocessing: thresholding often helped Tesseract but sometimes hurt cloud engines; Google Vision had the best baseline. No CER values in the abstract.
- **Santos, S. F. A. dos (2024). Optical character recognition, text analysis and key information extraction.** Master's thesis, Iscte – University Institute of Lisbon. http://hdl.handle.net/10071/32991 — [OCR

tools] [Related thesis] — OCR + KIE on SROIE receipts; regex remained effective for numbers and dates. No numbers in the abstract.

1.4 Inference optimisation (batching, quantization, latency)

- ★ **Juusti, M. (2026). Optimizing Transformer Inference Throughput for Batch Processing.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/147320 — [Async/batching] [A-3/A-4 analysis] — M2M-100 (1.2B) on an RTX 3090: batching strategy, runtime choice and CPU/GPU overlap combined gave 5.33x throughput with no measurable quality loss; largest gains from a specialised runtime and length-sorted batching; CPU/GPU overlap helped only when the runtime released the Python GIL during generation. Very close in method and magnitude to the thesis's 5.7x.
- **Fang, Z., Al Neama, M. (2025). Analyzing Factors Influencing Performance in LLM Inference Systems.** Master's thesis, Chalmers University of Technology. https://hdl.handle.net/20.500.12380/309294 — [Async/batching] [A-3/A-4 analysis] — Llama 3 and GPT-2 latency/throughput study: increasing batch size "significantly improves throughput" but can raise latency; allocating more GPU resources lowered overall GPU utilisation. No numbers in the abstract.
- **Samarasinghe Arachchige, S. (2025). Evaluating dynamic batching strategies for energy-efficient inference serving: A performance study.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/138117 — [Async/batching] — Triton dynamic batching with ResNet50/MobileNet: aggressive batching lowers energy and raises GPU utilisation but adds latency, especially for light models and low traffic. No numbers in the abstract.
- **Arte, E. (2024). Exploring the Effect of Quantization on Large Language Models for C++ Programming on Home Hardware.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/130282 — [Compression] — Code LLMs on home hardware with/without 4-bit quantization: 4-bit gave significantly faster output *when the unquantized model exceeded VRAM*, with no accuracy loss observed – the complementary case to the thesis (where fp32 Flan-T5-base already fits).
- **Barani, A. A., Mirzabeigi, A. (2025). Methods for Optimizing BERT Model on Edge Devices – Accelerating Biomedical NLP with Pruned and Quantized BERT Models.** Master's thesis, Chalmers University of Technology. https://hdl.handle.net/20.500.12380/310964 — [Compression] [A-3/A-4 analysis] — Structured pruning + ONNX dynamic INT8 on BERT-family NER: best setup cut mean CPU latency from 32.52 ms to 12.02 ms (2.6x) with F1 0.954 → 0.948 (integer kernels on CPU, unlike weight-only NF4 on GPU).
- **Dawli, M. (2026). Investigating Energy Consumption and Performance Trade-offs in Large Language Model Inference Using Quantization and Pruning.** Master's thesis (H2), KTH Royal Institute of Technology. http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-379106 — [Compression] [A-3/A-4 analysis] — FP32/FP16/INT8/INT4 + pruning on GPT-2 1.5B, DeepSeek-R1 1.5B/7B and Mistral 7B: FP16 gave "more than four times the energy saving of full-precision" with no accuracy loss; INT8/INT4 showed small accuracy drops attributed to implementation limits.
- **Tran, A. (2025). The impact of quantization on the inference performance and quality of large language models: A comparative study on code generation task.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/142491 — [Compression] — bitsandbytes 8/4-bit vs GPTQ 4-bit vs AWQ on Llama 3.1 8B / 3.2 3B (V100), measuring pass@k, memory, latency, throughput. No numbers in the abstract (read the results chapter for bitsandbytes latency).

1.5 Also verified, lower priority

- **Ylisiurunen, M. (2022). Extracting semi-structured information from receipts.** Master's thesis, Aalto University. https://aaltodoc.aalto.fi/handle/123456789/114516 — [Related thesis] — Reviews KIE models and fine-tunes LayoutXLM on SROIE + Finnish receipts; results mostly comparable to the original papers; data scarcity the main limit.
- **Mohan, A. (2023). Development of an End-to-End Pipeline for Custom Key-Value Extraction from Commercial Invoices.** M.Eng. thesis, Massachusetts Institute of Technology. https://hdl.handle.net/1721.1/150308 — [Related thesis] [OCR tools] — Image processing → OCR → custom algorithms → ML-based matching for distributor invoices. No numbers in the abstract.

- **Masa, B. A. (2023). Development of an Artificial Intelligence-based Solution for Document Processing Automation Using Machine Learning and NLP Techniques.** Master's thesis, University of Bologna. https://amslaurea.unibo.it/30008/ — [Related thesis] — Extracts invoice date, number, amount and supplier with sequence labelling; a custom NER model was competitive with little training data.
- **Krieger, F. F. A. (2024). Toward the Digitalization of Auditing: Applying Machine Learning for Information Extraction from Invoices.** PhD dissertation, Leuphana University Lüneburg. https://doi.org/10.48548/pubdata-741 — [Related thesis] — New ML architecture for invoice IE and pipeline design principles for audit use. Verified via the DataCite DOI record only (repository behind a bot wall).

No thesis on result/semantic caching for ML or LLM pipelines could be found and verified; cite caching from Section 5.3.

2. VLMs vs OCR pipelines

2.1 Does OCR text still help? Head-to-head accuracy evidence

- ★ **Borchmann, Ł. (2024). Notes on Applicability of GPT-4 to Document Understanding.** arXiv preprint arXiv:2405.18433. https://arxiv.org/abs/2405.18433 — [VLM vs OCR justification] [Evaluation metrics] — Evaluates text-only GPT-4 (given OCR text) and GPT-4 Vision Turbo (image only, or image + OCR text) on DocVQA, InfographicsVQA, SlideVQA and DUDE. Table 1 (ANLS; SlideVQA exact match): GPT-4V image only vs image + OCR text: DocVQA 84.6 → 87.4, InfoVQA 67.3 → 71.9, SlideVQA 55.1 → 57.3, DUDE 53.5 vs 53.3. Abstract: GPT-4V "performs well when one provides both text recognized by an external OCR engine" and the document images. Table 11: OCR engines used were Azure (DocVQA, DUDE), Tesseract (InfoVQA) and Amazon Textract (SlideVQA).
- ★ **Luo, C., Shen, Y., Zhu, Z., et al. (2024). LayoutLLM: Layout Instruction Tuning with Large Language Models for Document Understanding.** CVPR 2024, pp. 15630–15640. https://doi.org/10.1109/CVPR52733.2024.01480 (arXiv 2404.05225) — [VLM vs OCR justification] [Evaluation metrics] — Treats KIE as question answering scored with ANLS (same one-question-per-field formulation as the thesis). Table 1 zero-shot: text-only Llama2-7B-chat on plain OCR text SROIE 68.97 / CORD 47.70 vs Qwen-VL-7B 58.59 / 30.00 vs LLaVA-1.5-7B 3.83 / 18.06. Sec. 4.4: "Generally, the existing LLMs are better than MLLMs for zero-shot document VQA and VIE."
- ★ **Shpigel Nacson, M., Aberdam, A., Ganz, R., et al. (2025). DocVLM: Make Your VLM an Efficient Reader.** CVPR 2025, pp. 29005–29015. https://doi.org/10.1109/CVPR52734.2025.02701 (arXiv 2412.08746) — [VLM vs OCR justification] [A-3/A-4 analysis] — Table 1: unconstrained VLMs use about 16K (Qwen2-VL), 7K (LLaVA-OneVision) and 3.1K (InternVL2) visual tokens per document. With a constrained visual-token budget, adding OCR raises DocVQA for InternVL2 from 56.0% (256 tokens) to 86.6% (320 tokens incl. 64 OCR tokens) and for Qwen2-VL from 84.4% to 91.2% (320 tokens). Sec. 6: "allocating a small portion of tokens to OCR information consistently yields better results".
- ★ **Peer, D., Schöpf, P., Nebendahl, V., et al. (2024). ANLS* – A Universal Document Processing Metric for Generative Large Language Models.** arXiv preprint arXiv:2402.03848. https://arxiv.org/abs/2402.03848 — [VLM vs OCR justification] [Evaluation metrics] — Extends ANLS (τ = 0.5) to lists, dictionaries and None so a generative model's key-value output can be scored; "missing keys as well as hallucinated keys are penalized" (pip package `anls_star`). Table 4: on SROIE, gpt-4-turbo on OCR text (SFT prompt) 0.890 vs gpt-4-vision on images 0.834; DocVQA 0.770 vs 0.759; the authors note gpt-4-vision "is still behind our SFT solution which exploits an OCR scanner". Not universal: vision wins on Kleister Charity (0.751 vs 0.733) and VRDU Registration (0.665 vs 0.590).
- ★ **Borchmann, Ł., Pietruszka, M., Jaśkowski, W., et al. (2024). Arctic-TILT. Business Document Understanding at Sub-Billion Scale.** arXiv preprint arXiv:2408.04632. https://arxiv.org/abs/2408.04632 — [VLM vs OCR justification] — OCR-based (words + bounding boxes) encoder-decoder with "less than 1B parameters"; abstract: "accuracy on par with models 1000× its size on these use cases" and "It can be fine-tuned and deployed on a single 24GB GPU". Appendix A: "smaller models with an explicit OCR step still outperform" large vision-only models, and "even GPT-4 Vision benefits from the availability of OCR-recognized text". Table 3: DUDE 58.1 vs GPT-4Vt+OCR 53.9; but DocVQA 90.2 vs InternVL 2.0 Pro 95.1 and MMLongBench-Doc 25.8 vs GPT-4o 42.8 – not universally better.

- ★ **Jin, Y., Wang, Y., Zhong, Q., et al. (2025). Multi-Stage Field Extraction of Financial Documents with OCR and Compact Vision-Language Models.** arXiv preprint arXiv:2510.23066. https://arxiv.org/abs/2510.23066 — [VLM vs OCR justification] [A-3/A-4 analysis] — Compares PaddleOCR-based page filtering + a compact VLM (MiniCPM-o 2.6, 8B) with sending whole documents to Qwen2.5-VL-72B on 92 scanned financial reports. Abstract: "8.8 times higher field level accuracy", 0.7% of the GPU cost and 92.6% lower end-to-end latency; Table 1: field accuracy 80.87% vs 9.13%, 0.717 vs 9.656 s/page, 1,522 vs 11 docs/h per A100. The speed-up is attributed mostly to "the much faster OCR processing time to quickly filter out irrelevant pages" – parallels the thesis's OCR + MiniLM line filter.
- **Wang, W., Li, Y., Ou, Y., Zhang, Y. (2023). Layout and Task Aware Instruction Prompt for Zero-shot Document Image Question Answering (LATIN-Prompt).** arXiv preprint arXiv:2306.00526. https://arxiv.org/abs/2306.00526 — [VLM vs OCR justification] — Text-only LLMs read Azure OCR text with layout preserved through spaces and line breaks. Table 2 (DocVQA test ANLS): Claude 0.2298 → 0.8336 with the layout-aware prompt, on par with fine-tuned LayoutLMv3-LARGE (0.8337); ChatGPT-3.5 0.6866 → 0.8255.
- **Liu, X., Tang, W., Ni, X., et al. (2023). What Large Language Models Bring to Text-rich VQA?** arXiv preprint arXiv:2311.07306. https://arxiv.org/abs/2311.07306 — [VLM vs OCR justification] [OCR tools] — Training-free PaddleOCR + Vicuna-13B reaches DocVQA 0.4528 (Table 2, answer-containment accuracy) vs LLaVA 0.0514 and LLaVAR 0.0884; adding OCR output to MLLMs raised DocVQA from 0.1694 to 0.3299 (Sec. 4.4.3). Authors: "improving the quality of OCR results had a greater impact on VQA accuracy"; the bottleneck "may primarily lie in visual part".
- **Bhattacharyya, A., Tripathi, A., Das, U., et al. (2025). Information Extraction from Visually Rich Documents using LLM-based Organization of Documents into Independent Textual Segments (BLOCKIE).** ACL 2025 (Long Papers), pp. 17241–17256. https://doi.org/10.18653/v1/2025.acl-long.844 — [VLM vs OCR justification] [A-3/A-4 analysis] — OCR text + box LLM method that first splits documents into semantic blocks (parallel to the thesis's line filter). Table 1, Claude 3.5 Sonnet: SROIE 98.52 / CORD 98.83 F1 vs zero-shot Sonnet 91.37 / 88.92; Table 2, Qwen2.5-7B: CORD 87.72. Limitations: sequential LLM calls increase latency.
- **Liao, W., Wang, J., Li, H., et al. (2025). DocLayLLM: An Efficient Multi-modal Extension of Large Language Models for Text-rich Document Understanding.** CVPR 2025, pp. 4038–4049. https://doi.org/10.1109/CVPR52734.2025.00382 (arXiv 2408.15045) — [VLM vs OCR justification] — OCR text + visual patches into an LLM. Table 3 vs OCR-free models: SROIE 76.59 / FUNSD 80.28 vs TextMonkey 46.20 / 42.90 and DocKylin 25.50 / 58.70. Argues OCR-free methods "typically require high-resolution image inputs to recognize dense text", "significantly increasing computational demands".
- **Zhang, Y., Zhang, R., Gu, J., et al. (2023). LLaVAR: Enhanced Visual Instruction Tuning for Text-Rich Image Understanding.** arXiv preprint arXiv:2306.17107. https://arxiv.org/abs/2306.17107 — [VLM vs OCR justification] — Sec. 3: "running OCR at the original resolution … might recognize small fonts that are not visible by visual encoders like CLIP ViT"; DocVQA accuracy only 11.6% at 336^2 input (Table 2). The authors reject feeding OCR text at inference because it "significantly increases the computation (longer context lengths)" – an honest trade-off to discuss.
- **Lee, J., Park, S. (2025). Exploring OCR-augmented Generation for Bilingual VQA.** arXiv preprint arXiv:2510.02543. https://arxiv.org/abs/2510.02543 — [VLM vs OCR justification] — Abstract only (no numbers read): "OCR-extracted text significantly boosts performance across open source and commercial models" on Korean/English VQA.
- ★ **Shen, J., Yuan, P., Ghosh, A., et al. (2026). OCR or Not? Rethinking Document Information Extraction in the MLLMs Era with Real-World Large-Scale Datasets.** arXiv preprint arXiv:2603.02789. https://arxiv.org/abs/2603.02789 — [VLM vs OCR justification] [A-3/A-4 analysis] — **Main counter-evidence.** 11 MLLMs on about 1,000 real business documents: abstract concludes "OCR may not be necessary for powerful MLLMs"; best single result is image-only Gemini 1.5 Pro (mean F1 76.8). Yet in Table 1 OCR-only is at least as good as image-only for 7 of 11 models (count made from the table), e.g. Claude 3 Opus 50.1 → 70.1. Table 2: cloud latency about 1.4 s (Gemini 2.5 Flash) to about 7.0 s per page, about $0.001–$0.05 per page – a cloud baseline against which the local pipeline's latency and cost can be compared.

2.2 Benchmarks of VLM reading, KIE and hallucination

- ★ **Liu, Y., Li, Z., Huang, M., et al. (2024). OCRBench: On the Hidden Mystery of OCR in Large Multimodal Models.** Science China Information Sciences 67(12), 220102. https://doi.org/10.1007/s11432-024-4235-6 (arXiv 2305.07895) — [VLM vs OCR justification] [Evaluation metrics] — 1,000 QA pairs over 29 datasets incl. KIE (SROIE, FUNSD, POIE). Table 3: Gemini 659 and GPT-4V 645 of 1000; KIE 134 and 160 of 200; handwritten math 8 and 9 of 100; GPT-4V refused 84 images. Sec. 3.2: on handwriting "LMMs exhibit a 51.9% lower performance compared to the supervised state-of-the-art model"; Sec. 5: "significant gaps persist compared to domain-specific supervised methods".
- **Fu, L., Kuang, Z., Song, J., et al. (2025). OCRBench v2: An Improved Benchmark for Evaluating Large Multimodal Models on Visual Text Localization and Reasoning.** NeurIPS 2025 Datasets & Benchmarks Track. https://arxiv.org/abs/2501.00321 — [VLM vs OCR justification] [A-3/A-4 analysis] — 10,000 human-verified QA pairs, 23 sub-tasks; most LMMs score below 50/100. Text spotting: GPT-4o 0, Claude-3.5-Sonnet 1.3, Gemini-1.5-Pro 13.5 vs specialist TESTR 51.8 – i.e. LMMs cannot reliably say *where* text is (no audit trail). Document parsing: GOT 53.9 vs GPT-4o 35.9. "LMMs still struggle to parse text into structured formats".
- ★ **Yang, Z., Tang, J., Li, Z., et al. (2025). CC-OCR: A Comprehensive and Challenging OCR Benchmark for Evaluating Large Multimodal Models in Literacy.** ICCV 2025, pp. 21744–21754. https://doi.org/10.1109/ICCV51701.2025.02019 (arXiv 2412.02210) — [VLM vs OCR justification] [Evaluation metrics] — 7,058 images, 4 tracks incl. KIE (2,008 images) with field-level F1 = 2TP/(2TP+FP+FN) where one wrong character fails the field. Total KIE F1 (Table 5): Qwen2-VL-72B 71.76, Gemini-1.5-Pro 67.28, Claude-3.5-Sonnet 64.58, GPT-4o 63.45. Reports "hallucinations in the form of repeated text" (TextMonkey repetition rate 33.93%, Table 8). Caveat: generalist LMMs beat specialised OCR-LMMs on multi-scene OCR; do not compare these F1s directly with supervised SROIE scores (different protocols).
- **Shi, Y., Peng, D., Liao, W., et al. (2023). Exploring OCR Capabilities of GPT-4V(ision): A Quantitative and In-depth Evaluation.** arXiv preprint arXiv:2310.16809. https://arxiv.org/abs/2310.16809 — [VLM vs OCR justification] [A-3/A-4 analysis] — Handwriting (IAM line, Table 4): GPT-4V WER 33.42% / CER 13.75% vs SOTA 21.47% / 6.52%; FUNSD entity labelling F1 34.51% (Table 8); Sec. 2.5: GPT-4V "tends to generate new keys" not in the document; Sec. 2.2: "significant hallucinations in Chinese text recognition". Abstract: "we affirm the necessity and continued research value of specialized OCR models."
- ★ **He, Z., Zhang, C., Wu, Z., et al. (2025). Seeing is Believing? Mitigating OCR Hallucinations in Multimodal Large Language Models.** NeurIPS 2025. https://arxiv.org/abs/2506.20168 — [VLM vs OCR justification] [A-3/A-4 analysis] — Introduces KIE-HVQA (degraded receipts, ID cards, OCRBench-KIE items). Table 1 overall: GPT-4o 30.21, Claude-3.5-Sonnet 24.25, Gemini-2.5-pro 35.03, Qwen2.5-VL-72B 21.53; Fig. 1: Qwen2.5-VL-7B "may experience hallucinations, identifying values not present in the image". Abstract: the authors' RL-trained 7B model "achieves a 22% absolute improvement in hallucination-free accuracy over GPT-4o".
- ★ **Ouyang, L., Qu, Y., Zhou, H., et al. (2025). OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations.** CVPR 2025, pp. 24838–24848. https://doi.org/10.1109/CVPR52734.2025.02313 (arXiv 2412.07626) — [VLM vs OCR justification] [OCR tools] — Overall edit distance on English pages (Table 2, lower is better): pipeline MinerU 0.15 vs GPT-4o 0.233, Qwen2-VL-72B 0.252, InternVL2-76B 0.44; newspapers text (Table 3): MinerU 0.171 vs GPT-4o 0.751. "General VLMs often lag behind specialized pipelines and expert models on standard documents", and some VLMs "generate hallucinated information". Counterpoint: VLMs are better on handwritten notes (Qwen2-VL-72B 0.168 vs MinerU 0.984).
- ★ **Livathinos, N., Auer, C., Lysak, M., et al. (2025). Docling: An Efficient Open-Source Toolkit for AI-driven Document Conversion.** AAAI-25 Workshop on Open-Source AI for Mainstream Use; arXiv:2501.17887. https://arxiv.org/abs/2501.17887 — [VLM vs OCR justification] [OCR tools] — Sec. 2 on VLM-based conversion tools: "they are prone to hallucinations", "these models demand substantial computational resources, making the conversion process expensive", and they are "typically offered as SaaS, with compute-intensive operations performed remotely in the cloud"; Intro: they "cannot be run easily on local hardware". Sec. 5.4: Docling 3.1 s/page on x86 CPU and 0.49 s/page on an L4 GPU; Marker "over 16 sec/page (x86 CPU)".

- **Blecher, L., Cucurull, G., Scialom, T., Stojnic, R. (2024). Nougat: Neural Optical Understanding for Academic Documents.** ICLR 2024. https://openreview.net/forum?id=fUtxNAKpdV (arXiv 2308.13418) — [VLM vs OCR justification] [OCR tools] — OCR-free 350M image-to-markup model. Sec. 5.4: the model "degenerates into repeating the same sentence" on 1.5% of test pages (more often out of domain). Sec. 5.5: on an A10G (24 GB) a batch of 6 pages takes 19.5 s on average, while the rule-based GROBID handles 10.6 PDF/s.
- **Mathew, M., Karatzas, D., Jawahar, C. V. (2021). DocVQA: A Dataset for VQA on Document Images.** WACV 2021. https://doi.org/10.1109/WACV48630.2021.00225 (arXiv 2007.00398) — [VLM vs OCR justification] [Evaluation metrics] — 50,000 questions on 12,000+ document images, scored with ANLS and accuracy. Table 1: answers are "a substring of the serialized OCR output in more than 85% of the cases" (87.00 on test) – the task is largely extractive over OCR text. Humans 0.981 ANLS; best fine-tuned BERT-large baseline 0.665 ANLS (Table 3).
- **Bai, Z., Wang, P., Xiao, T., et al. (2024). Hallucination of Multimodal Large Language Models: A Survey.** arXiv preprint arXiv:2404.18930. https://arxiv.org/abs/2404.18930 — [VLM vs OCR justification] — Definition to cite: hallucination occurs when "the generated text response does not align with the corresponding visual content" (Sec. 2.3). No OCR-specific numbers; pair with KIE-HVQA and CC-OCR.
- **Ye, J., Hu, A., Xu, H., et al. (2023). UReader: Universal OCR-free Visually-situated Language Understanding with Multimodal Large Language Model.** Findings of EMNLP 2023, pp. 2841–2858. https://doi.org/10.18653/v1/2023.findings-emnlp.187 — [VLM vs OCR justification] — Representative OCR-free MLLM; needs shape-adaptive cropping to read high-resolution images; state of the art among OCR-free models on 8 of 10 tasks. DocOwl2 (Table 2) lists it at about 841 visual tokens per page with DocVQA ANLS 65.4.
- **Liu, Y., Yang, B., Liu, Q., et al. (2024). TextMonkey: An OCR-Free Large Multimodal Model for Understanding Document.** arXiv preprint arXiv:2403.04473. https://arxiv.org/abs/2403.04473 — [VLM vs OCR justification] — Representative OCR-free LMM (OCRBench 561; abstract reports +2.8% on KIE). DocLayLLM's Table 3 reports its zero-shot SROIE at only 46.20, and CC-OCR measures a 33.93% repetition rate.
- **OpenAI (2026). Images and vision – API guide.** Official documentation, accessed 2026-10-20. https://developers.openai.com/api/docs/guides/images-vision — [VLM vs OCR justification] — Images are billed as input tokens; for tile-based models (gpt-4o, gpt-4.1) 85 base tokens + 170 per 512 px tile at high detail. The vendor's own "Limitations" list: "may not perform optimally when handling images with text of non-Latin alphabets", "may misinterpret rotated or upside-down text", "struggles with tasks requiring precise spatial localization", "may generate incorrect descriptions or captions in certain scenarios". (Living document; quote with access date.)

2.3 Compute, memory and visual-token cost of (small) VLMs

- ★ **Marafioti, A., Zohar, O., Farré, M., et al. (2025). SmolVLM: Redefining small and efficient multimodal models.** arXiv preprint arXiv:2504.05299. https://arxiv.org/abs/2504.05299 — [VLM vs OCR justification] [Compression] [A-3/A-4 analysis] — Abstract: the 256M model "uses less than 1GB GPU memory during inference". Table 1 batch-1 GPU RAM 0.8 / 1.2 / 4.9 GB (256M / 500M / 2.2B) with DocVQA 58.3% / 70.5% / 80.0%. Sec. 4.3: "Qwen2VL-2B requires 13.7GB VRAM and InternVL2-2B requires 10.5GB VRAM"; Moondream2 3.9 GB. Shows the accuracy/memory trade-off: the sub-1 GB models are much weaker on documents.
- **Marafioti, A., Noyan, M., Farré, M., et al. (2024). SmolVLM – small yet mighty Vision Language Model.** Hugging Face blog (non-peer-reviewed), 26 Nov 2024. https://huggingface.co/blog/smolvlm — [VLM vs OCR justification] — Minimum GPU RAM table: SmolVLM 5.02 GB, Qwen2-VL 2B 13.70, InternVL2 2B 10.52, PaliGemma 3B 448px 6.72, moondream2 3.87; one test prompt + image is "1.2k tokens, whereas Qwen2-VL uses 16k tokens". All except moondream2 exceed a 4 GB GPU.
- ★ **Qwen Team (2024). Qwen2-VL-2B-Instruct.** Hugging Face model card. https://huggingface.co/Qwen/Qwen2-VL-2B-Instruct — [VLM vs OCR justification] [Compression] — "The default range for the number of visual tokens per image in the model is 4-16384"; example `min_pixels = 256*28*28`, `max_pixels = 1280*28*28` gives "a token count range of 256-1280, to balance speed and memory

usage" (≈ one token per 28×28 px); "higher resolutions can enhance performance at the cost of more computation". Reports DocVQA 90.1. (Own arithmetic, not from the card: an A4 page at 150 DPI ≈ 2.17 MP ≈ 2,770 visual tokens; bf16 weights alone ≈ 2 × 2B ≈ 4 GB by the HF rule in Section 5.)

- ★ **Beyer, L., Steiner, A., Pinto, A. S., et al. (2024). PaliGemma: A versatile 3B VLM for transfer.** arXiv preprint arXiv:2407.07726. https://arxiv.org/abs/2407.07726 — [VLM vs OCR justification] [Compression] — SigLIP 400M + Gemma 2B ("less than 3 B total parameters"). Sec. 3.1: resolutions 224/448/896 px give "respectively 256, 1024, or 4096 tokens". Table 1: DocVQA 43.7 (224 px) → 78.0 (448 px) → 84.8 (896 px) – document accuracy requires high resolution, i.e. many visual tokens.
- **Steiner, A., Pinto, A. S., Tschannen, M., et al. (2024). PaliGemma 2: A Family of Versatile VLMs for Transfer.** arXiv preprint arXiv:2412.03555. https://arxiv.org/abs/2412.03555 — [VLM vs OCR justification] [Compression] — 3B/10B/28B; 224/448/896 px "yielding 256/1024/ 4096 tokens"; Sec. 4.1.1: going to 448 px costs "4.6× more FLOPs". Sec. 4.9 / Table 10: 3B model memory 12.1 GB (JAX, F32) vs 4.0 GB (gemma.cpp, quantized, CPU).
- **Yao, Y., Yu, T., Zhang, A., et al. (2024). MiniCPM-V: A GPT-4V Level MLLM on Your Phone.** arXiv preprint arXiv:2408.01800. https://arxiv.org/abs/2408.01800 — [VLM vs OCR justification] [Compression] — Up to 1.8 MP (e.g. 1344×1344); "each slice is encoded into 1,024 tokens, where 10 slices can yield over 10k tokens", compressed to 64/96 per slice. Sec. 5.2: the 8B fp16 model "typically demands 16~17G memory", Q4_K_M reduces this "to around 5G". Sec. 5.4: "the current computation bottleneck primarily stems from LLM prefilling, which mainly involves encoding image and text tokens". The MiniCPM-V 2.6 card (https://huggingface.co/openbmb/MiniCPM-V-2_6) offers int4 "for lower GPU memory (7GB) usage".
- **Chen, Z., Wang, W., Tian, H., et al. (2024). How Far Are We to GPT-4V? Closing the Gap to Commercial Multimodal Models with Open-Source Suites (InternVL 1.5).** arXiv preprint arXiv:2404.16821. https://arxiv.org/abs/2404.16821 — [VLM vs OCR justification] [Compression] — Dynamic tiling with "tiles ranging from 1 to 40 of 448×448 pixels"; "a 448×448 image is represented by 256 visual tokens"; up to "10,496 visual tokens" at test. InternVL2-2B card (https://huggingface.co/OpenGVLab/InternVL2-2B): InternViT-300M + internlm2-chat-1.8B, DocVQA 86.9.
- **Abdin, M., Aneja, J., Awadalla, H., et al. (2024). Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone.** arXiv preprint arXiv:2404.14219. https://arxiv.org/abs/2404.14219 — [VLM vs OCR justification] — "phi-3.5-Vision, a 4.2 billion parameter model"; CLIP ViT-L/14 with dynamic cropping; TextVQA 72.0. Model card (https://huggingface.co/microsoft/Phi-3.5-vision-instruct) lists tested GPUs as A100, A6000 and H100.
- **Wu, Z., Chen, X., Pan, Z., et al. (2024). DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding.** arXiv preprint arXiv:2412.10302. https://arxiv.org/abs/2412.10302 — [VLM vs OCR justification] [Compression] — Tiles of 384×384 compressed to 196 tokens each (up to 9 tiles + thumbnail); conclusion: "our 3B, 16B and 27B models can be deployed on a single GPU with 10 GB, 40GB and 80GB memory respectively". Tiny: DocVQA 88.9.
- **Granite Vision Team (IBM) (2025). Granite Vision: a lightweight, open-source multimodal model for enterprise Intelligence.** arXiv preprint arXiv:2502.09927. https://arxiv.org/abs/2502.09927 — [VLM vs OCR justification] — Document-focused VLM of "approximately 3 billion parameters", up to ten 384×384 tiles plus a global image; DocVQA 0.88 (Table 1). No memory/latency figures given.
- **Bai, S., Chen, K., Liu, X., et al. (Qwen Team) (2025). Qwen2.5-VL Technical Report.** arXiv preprint arXiv:2502.13923. https://arxiv.org/abs/2502.13923 — [Compression] — 3B/7B/72B; images resized to multiples of 28 px, 14 px patches with 2×2 merge (≈ one token per 28×28 px); 3B: DocVQA 93.9, OCRBench 797 (Table 5). No memory/latency numbers.
- **Xiao, B., Wu, H., Xu, W., et al. (2024). Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks.** CVPR 2024, pp. 4818–4829. https://arxiv.org/abs/2311.06242 ; model card https://huggingface.co/microsoft/Florence-2-base — [VLM vs OCR justification] — **Counter-example on size:** Florence-2-B has "232 million parameters" (≈ Flan-T5-base) and Florence-2-L 771M; 768×768 input. The model card lists only fixed task prompts (`<OCR>`, `<OCR_WITH_REGION>`, captioning, detection) – no free-form document QA / field extraction, so it can replace Tesseract but not the extraction step. Caveat: the text says TextVQA 81.5 while Table 5 says 73.5 – check the PDF before quoting.

- **Wei, H., Liu, C., Chen, J., et al. (2024). General OCR Theory: Towards OCR-2.0 via a Unified End-to-end Model (GOT-OCR2.0).** arXiv preprint arXiv:2409.01704. https://arxiv.org/abs/2409.01704 — [VLM vs OCR justification] [OCR tools] [Compression] — Dedicated 580M OCR model (80M encoder + Qwen-0.5B decoder). Sec. 3.3.1: "1024×1024 optical pixels to 256 image tokens", "easier to deploy on a consumer-grade GPU with 4G memory" (claim, not measured). Table 1 English document OCR edit distance 0.035 vs Qwen-VL-Max 0.057 and InternVL-ChatV1.5 (26B) 0.393. Authors: "LVLM often enjoys billions of parameters, leading to the post-training and deployment costs being too high." Argues for a dedicated OCR stage, even if neural.
- **Nassar, A., Marafioti, A., Omenetti, M., et al. (2025). SmolDocling: An ultra-compact vision-language model for end-to-end multi-modal document conversion.** ICCV 2025; arXiv:2503.11576. https://arxiv.org/abs/2503.11576 — [VLM vs OCR justification] [Compression] — **Counter-evidence:** a 256M VLM; Sec. 5.1: "page conversion time of 0.35 seconds" on an A100 with vLLM "while occupying only 0.489 GB of VRAM"; 64 visual tokens per 512×512 patch. Still shows "endless repetition of tokens" (Sec. 5.3). Converts pages to markup; it does not do key-field extraction.
- ★ **Hu, A., Xu, H., Zhang, L., et al. (2025). mPLUG-DocOwl2: High-resolution Compressing for OCR-free Multi-page Document Understanding.** ACL 2025 (Long Papers), pp. 5817–5834. https://doi.org/10.18653/v1/2025.acl-long.291 — [VLM vs OCR justification] [A-3/A-4 analysis] — Visual tokens per page (Table 2): IXC2.5 ~5,118, InternVL2 ~3,133, DocOwl1.5 ~1,698, TextMonkey 768, DocOwl2 324; first-token latency (Table 3) 3.73 s (IXC2.5), 0.94 s (InternVL2), 0.26 s (DocOwl2). Compression costs accuracy: DocVQA ANLS 80.7 (DocOwl2) vs 91.6 (InternVL2). Abstract cites "excessive GPU memory and slower inference times" of high-resolution OCR-free models.
- **Yu, Y.-Q., Liao, M., Wu, J., et al. (2024). TextHawk: Exploring Efficient Fine-Grained Perception of Multimodal Large Language Models.** arXiv preprint arXiv:2404.09204. https://arxiv.org/abs/2404.09204 — [VLM vs OCR justification] — Sec. 1: "Document images typically have higher resolution and higher information density than natural images"; Sec. 3.2.2: a document on a 5×4 sub-image grid "will consume up to 5,120 tokens", hence a 16× token compression.
- **Chen, L., Zhao, H., Liu, T., et al. (2024). An Image is Worth 1/2 Tokens After Layer 2: Plug-and-Play Inference Acceleration for Large Vision-Language Models (FastV).** ECCV 2024; arXiv:2403.06764. https://arxiv.org/abs/2403.06764 — [Compression] [VLM vs OCR justification] — "LLAVA process 336x336 images into 576 tokens", 672×672 gives "2304 tokens"; Fig. 3: "In average, image tokens take up most of the input tokens (64%)".
- **Jin, Y., Li, J., Liu, Y., et al. (2025). Efficient Multimodal Large Language Models: A Survey.** Visual Intelligence 3, 27. https://doi.org/10.1007/s44267-025-00099-6 (arXiv 2405.10739) — [VLM vs OCR justification] [A-3/A-4 analysis] — Sec. 1: "performing inference with LLaVA-1.5 and a Vicuna-13B LLM backbone requires 18.2T FLOPS and 41.6G of memory usage"; higher resolution means more visual tokens, a burden "primarily due to the quadratic scaling of computational costs with the number of input tokens"; high requirements "make it challenging to run them on edge devices".
- ★ **Poznanski, J., Rangapur, A., Borchardt, J., et al. (2025). olmOCR: Unlocking Trillions of Tokens in PDFs with Vision Language Models.** arXiv preprint arXiv:2502.18443. https://arxiv.org/abs/2502.18443 — [VLM vs OCR justification] [A-3/A-4 analysis] — Cost per million pages (Table 6): olmOCR 7B VLM $176 on an L40S; GPT-4o API $12,480 ($6,240 batch); MinerU $596; Marker $1,484. Footnote: GPT-4o mini "produced excessive hallucinations".

2.4 Cost, energy, control and reproducibility

- ★ **Luccioni, S., Jernite, Y., Strubell, E. (2024). Power Hungry Processing: Watts Driving the Cost of AI Deployment?** ACM FAccT '24, pp. 85–99. https://doi.org/10.1145/3630106.3658542 (arXiv 2311.16863) — [VLM vs OCR justification] [A-3/A-4 analysis] — Measures energy and carbon for 1,000 inferences across many tasks and models. Abstract: multi-purpose generative architectures are "orders of magnitude more expensive than task-specific systems". Sec. 4.1: text classification 0.002 kWh vs image generation 2.9 kWh per 1,000 inferences ("a factor of over 1450"). Table 3: Flan-T5-base 0.026 kWh / 3.67 g CO2eq vs Flan-T5-xxl 0.083 kWh / 11.48 g per 1,000 inferences; Sec. 5: a BERT-based extractive QA model emits 0.70 g vs 2.36 g for Flan-T5-base.

- **Maiya, A. S. (2025). OnPrem.LLM: A Privacy-Conscious Document Intelligence Toolkit.** arXiv preprint arXiv:2505.07672. https://arxiv.org/abs/2505.07672 — [VLM vs OCR justification] — Sec. 1: organisations in "defense, healthcare, finance, and law" face requirements that "frequently prohibit the use of general-purpose external services"; options are small local models "with reduced performance", self-hosting large models "at significant infrastructure cost", or hybrid pipelines, each with "trade-offs in accuracy, scalability, latency, and maintainability". Its ingest module uses OCR for image-based text.
- **Atil, B., Aykent, S., Chittams, A., et al. (2024). Non-Determinism of "Deterministic" LLM Settings.** arXiv preprint arXiv:2408.04667. https://arxiv.org/abs/2408.04667 — [VLM vs OCR justification] — Five LLMs, eight tasks, 10 runs with settings "configured to be deterministic": "accuracy variations up to 15% across naturally occurring runs", best-to-worst gap "up to 70%"; "none of the LLMs consistently delivers repeatable accuracy across all tasks, much less identical output strings." Supports the control/reproducibility argument for a local, fixed-weight model.
- **Chen, L., Zaharia, M., Zou, J. (2024). How Is ChatGPT's Behavior Changing Over Time?** Harvard Data Science Review 6(2). https://doi.org/10.1162/99608f92.5317da47 (arXiv 2307.09009) — [VLM vs OCR justification] — Same API model name, different behaviour over time: GPT-4 prime-vs-composite accuracy 84% (March 2023) vs 51% (June 2023); more formatting mistakes in generated code in June. Supports the argument that hosted models drift, whereas a pinned local model does not.

3. OCR engines

- ★ **Hegghammer, T. (2022). OCR with Tesseract, Amazon Textract, and Google Document AI: a benchmarking experiment.** Journal of Computational Social Science 5(1), 861–882. https://doi.org/10.1007/s42001-021-00149-1 (open-access copy: https://ora.ox.ac.uk/objects/uuid:0f922084-1e7a-425a-92b0-7cbc7aba7626) — [OCR tools] [A-3/A-4 analysis] — Benchmarks the three engines on 322 English book scans and 100 Arabic articles, each with 43 artificial noise variants (18,568 documents). Results: "Document AI had consistently lower error rates, with Textract coming in a close second, and Tesseract last … Tesseract was significantly more sensitive to noise … Only on noise-free documents in English could Tesseract compete." Speed: cloud APIs "approximately 10–15 s per page"; Tesseract "17 s per page for Arabic and 2 seconds per page for English" (12-core CPU). Downstream classification shows "only a small negative effect … up to a threshold of around 20% OCR error, after which accuracy plummets".
- ★ **Hemmer, A., Coustaty, M., Bartolo, N., et al. (2024). Confidence-Aware Document OCR Error Detection.** arXiv preprint arXiv:2409.04117 (listed as a DAS 2024 paper on the IAPR DAS site; no proceedings DOI confirmed). https://arxiv.org/abs/2409.04117 — [OCR tools] [A-3/A-4 analysis] — Compares cloud and open-source OCR on receipts and proposes using OCR confidence scores to detect errors. Table 2, receipt CER (%) on SROIE: AWS 2.3, Azure 1.7, Google 2.8, docTR 4.0, EasyOCR 12.0, PaddleOCR 5.3 (CORD: 1.3 / 1.3 / 5.3 / 3.2 / 17.4 / 4.3). "Among the open-source OCRs, DocTR exhibits the lowest Character Error Rate, followed by PaddleOCR and EasyOCR"; errors cluster on punctuation and special characters (critical for totals and invoice numbers). Tesseract not tested. Supports using OCR confidence as a routing signal.
- ★ **Carlson, J., Bryan, T., Dell, M. (2024). Efficient OCR for Building a Diverse Digital History.** Proceedings of ACL 2024 (Vol. 1: Long Papers), pp. 8105–8115. https://doi.org/10.18653/v1/2024.acl-long.440 — [OCR tools] [A-3/A-4 analysis] — Table 1 compares off-the-shelf engines on English newspaper lines: CER Google Cloud Vision 0.005, Tesseract (Best) 0.106, EasyOCR CRNN 0.170, PaddleOCR SVTR 0.304; CPU throughput (four 2.2 GHz cores) Tesseract 4.47, EasyOCR 19.80, PaddleOCR 13.56 lines/s; 1.4M / 3.8M / 11M parameters. Caveat: historical newspapers, not invoices.
- **Nagayi, M., Khan, A., Frank, T., et al. (2025). Evaluating OCR performance on food packaging labels in South Africa.** arXiv preprint arXiv:2510.03570. https://arxiv.org/abs/2510.03570 — [OCR tools] — Tesseract, EasyOCR, PaddleOCR and TrOCR on 1,628 images (113 hand-transcribed). Table 3: "Tesseract was the fastest system, with an average of 0.58 seconds per image" (CPU); EasyOCR 0.81 s and TrOCR 2.20 s on an RTX 3070; PaddleOCR 6.24 s on CPU. Tesseract had the "lowest CER (0.912)" but lowest coverage (79.66%). Hard domain (stylised text), so absolute CERs are high.

- **Livathinos et al. (2025), Docling (AAAI-25 workshop)** – full entry in Section 2.2 — Sec. 5.4 gives EasyOCR timing: "1.6 s on the L4 GPU, 13 s on the x86 CPU and 5 s on the M3 Max SoC" per page; EasyOCR is "the biggest compute expense in the pipeline".
- **Auer, C., Lysak, M., Nassar, A., et al. (2024). Docling Technical Report.** arXiv preprint arXiv:2408.09869. https://arxiv.org/abs/2408.09869 — [OCR tools] — Describes IBM's open-source local document-conversion pipeline (layout model + TableFormer + optional OCR). Sec. 3.2: relies on EasyOCR, which "runs fairly slow on CPU (upwards of 30 seconds per page)". Table 1 (OCR disabled, 225 pages): 1.27 pages/s on Apple M3 Max (4 threads), 0.60 pages/s on Intel Xeon E5-2690 (4 threads), ~6.2 GB memory with the native backend.
- ★ **Li, C., Liu, W., Guo, R., et al. (2022). PP-OCRv3: More Attempts for the Improvement of Ultra Lightweight OCR System.** arXiv preprint arXiv:2206.03001. https://arxiv.org/abs/2206.03001 — [OCR tools] [A-3/A-4 analysis] — Table 5 (Hmean % / size MB / CPU ms / T4 GPU ms): PP-OCR mobile 50.3 / 8.1 / 356 / 116; PP-OCR server 57.0 / 155.1 / 1056 / 200; PP-OCRv2 57.6 / 11.6 / 330 / 111; PP-OCRv3 62.9 / 15.6 / 331 / 87 (Intel Xeon Gold 6148). "The Hmean of PP-OCRv3 is 5.3% higher than that of PP-OCRv2 with the same inference cost on CPU … 22% faster than PP-OCRv2 on T4 GPU."
- **Du, Y., Li, C., Guo, R., et al. (2020). PP-OCR: A Practical Ultra Lightweight OCR System.** arXiv preprint arXiv:2009.09941. https://arxiv.org/abs/2009.09941 — [OCR tools] — Detector + direction classifier + recognizer with a "bag of strategies" for small size. Abstract: "The overall model size of the PP-OCR is only 3.5M for recognizing 6622 Chinese characters and 2.8M for recognizing 63 alphanumeric symbols."
- **Du, Y., Li, C., Guo, R., et al. (2021). PP-OCRv2: Bag of Tricks for Ultra Lightweight OCR System.** arXiv preprint arXiv:2109.03144. https://arxiv.org/abs/2109.03144 — [OCR tools] — Adds collaborative mutual learning, CopyPaste augmentation, PP-LCNet backbone, U-DML and enhanced CTC loss; abstract: precision "7% higher" than PP-OCR "under the same inference cost". (Absolute figures: see PP-OCRv3 Table 5.)
- **Mindee (2021). docTR: Document Text Recognition.** GitHub repository and documentation (no peer-reviewed paper; the repository's own citation is @misc). https://github.com/mindee/doctr ; https://mindee.github.io/doctr/latest/using_doctr/using_models.html — [OCR tools] — PyTorch OCR library (Apache-2.0) with DBNet/LinkNet/FAST detectors and CRNN/SAR/MASTER/ViTSTR/PARSeq recognizers. Docs table (end-to-end OCR, recall/precision %) on CORD: docTR db_resnet50+master 85.84 / 80.07 vs AWS Textract 87.50 / 66.00 vs Azure Form Recognizer 89.62 / 88.93. Recognition speed (batch 64, i7-11800H CPU): crnn_mobilenet_v3_small 0.05 s/it vs master 17.6 s/it.
- **JaidedAI (n.d.). EasyOCR.** GitHub repository. https://github.com/JaidedAI/EasyOCR — [OCR tools] — "Ready-to-use OCR with 80+ supported languages" (Apache-2.0); "Detection execution uses the CRAFT algorithm" and "The recognition model is a CRNN". Cite as the tool reference; underlying methods: Baek et al. 2019 (CRAFT, CVPR 2019, https://doi.org/10.1109/CVPR.2019.00959) and Shi et al. 2017 (CRNN, IEEE TPAMI 39(11), https://doi.org/10.1109/TPAMI.2016.2646371) – bibliographic records and abstracts verified.
- **Li, M., Lv, T., Chen, J., et al. (2023). TrOCR: Transformer-Based Optical Character Recognition with Pre-trained Models.** Proceedings of AAAI 37(11), 13094–13102. https://doi.org/10.1609/aaai.v37i11.26538 — [OCR tools] — End-to-end Transformer OCR (image Transformer encoder + text Transformer decoder); abstract claims it "outperforms the current state-of-the-art models on the printed, handwritten and scene text recognition tasks". Works on single text lines, so it still needs a separate text detector.
- ★ **Tesseract OCR contributors (n.d.). Improving the quality of the output (tessdoc).** Official documentation. https://tesseract-ocr.github.io/tessdoc/ImproveQuality.html — [OCR tools] [A-3/A-4 analysis] — "Tesseract works best on images which have a DPI of at least 300 dpi"; "The quality of Tesseract's line segmentation reduces significantly if a page is too skewed"; built-in Otsu binarisation "can be suboptimal, particularly if the page background is of uneven darkness". Lists page segmentation modes (PSM 3 default "Fully automatic page segmentation, but no OSD", PSM 4, 6, 11 sparse text). Supports the preprocessing/PSM choices in Architecture A.
- ★ **van Strien, D., Beelen, K., Ardanuy, M. C., et al. (2020). Assessing the Impact of OCR Quality on Downstream NLP Tasks.** Proceedings of ICAART 2020, Vol. 1, pp. 484–496. https://doi.org/10.5220/0009169004840496 — [A-3/A-4 analysis] — Measures six downstream NLP tasks on OCR'd vs human-corrected text across four OCR-quality bands. Sec. 4.1.2: person entities "0.87 and 0.63 average f-score in quality bands 1 and 4"; GPE entities "greatly affected by noisy OCR'd text (0.76 and 0.54 …)". Evidence that OCR errors propagate to extraction.

- **Hamdi, A., Jean-Caurant, A., Sidère, N., et al. (2020). Assessing and Minimizing the Impact of OCR Quality on Named Entity Recognition.** TPDL 2020, LNCS 12246, pp. 87–101. https://doi.org/10.1007/978-3-030-54956-5_7 — [A-3/A-4 analysis] — NER on synthetic OCR noise of several levels/types plus real newspaper OCR; abstract reports "an increase of around 11% points in terms of F1-measure over the best-known results". The authors' slides (not the paper) state BLSTM NER stays satisfactory below "~15%" CER and "~30%" WER. Extended version: Natural Language Engineering 29(2):425–448 (2023), https://doi.org/10.1017/S1351324922000110 (paywalled, not read).
- **Zhang, J., Zhang, Q., Wang, B., et al. (2025). OCR Hinders RAG: Evaluating the Cascading Impact of OCR on Retrieval-Augmented Generation (OHRBench).** ICCV 2025, pp. 17443–17453. https://doi.org/10.1109/ICCV51701.2025.01620 (arXiv 2412.02592) — [A-3/A-4 analysis] [VLM vs OCR justification] — Benchmarks OCR solutions as the first stage of RAG on 8,561 document images. Sec. 4.2: "Even the best solutions experience a 14% (5 F1-score) drop in the overall stage evaluation" relative to ground-truth text; image-based VisRAG (40.2) does not beat OCR-based RAG (40.6), ground truth 46.8 (Table S4). Shows both that OCR errors propagate and that skipping OCR does not remove the loss.
- **Bubula, M., Baierer, K., Lehmann, J., et al. (2025). How Scalable is Quality Assessment of Text Recognition? A Combination of Ground Truth and Confidence Scores.** Anthology of Computers and the Humanities, Vol. 3. https://doi.org/10.63744/GR59c1iXu6Wj — [OCR tools] [A-3/A-4 analysis] — Berlin State Library study relating word confidence to word error rate for several OCR engines on almost 5 million historical pages (1456–2000); abstract: confidence "can serve as a useful indicator of OCR quality". No numbers read; supports (with Hemmer et al. 2024) using OCR confidence as a routing signal.

4. LLM-based invoice and receipt extraction

- ★ **Anvari, Z. (2026). From Pixels to Pairs: A Comprehensive Benchmark of LLM-Driven Key–Value Extraction in Noisy Document Settings.** arXiv preprint arXiv:2609.17538 (v2, Sep 2026). https://arxiv.org/abs/2609.17538 — [A-3/A-4 analysis] [Evaluation metrics] — Closest match to the thesis setup: five open instruction-tuned 2B–8B LLMs (Gemma-2B/7B, Mistral-7B, Qwen2.5-7B, LLaMA-3-8B) extract key-value pairs from gold text vs Tesseract, EasyOCR and PaddleOCR output (136 configurations, 17,688 document-level inferences), scored by Key Recall, Exact Match and Value F1. Table 5 (SROIE, zero-shot) Qwen2.5-7B Value F1: Gold 0.8930, PaddleOCR 0.8367, EasyOCR 0.6568, Tesseract 0.6742. Finding: "OCR corruption often preserves enough information to identify the field type while altering the value itself"; model rankings change between gold and OCR text.
- ★ **Ma, Y., Cao, Y., Hong, Y., Sun, A. (2023). Large Language Model Is Not a Good Few-shot Information Extractor, but a Good Reranker for Hard Samples!** Findings of EMNLP 2023, pp. 10572–10601. https://doi.org/10.18653/v1/2023.findings-emnlp.710 — [A-3/A-4 analysis] [VLM vs OCR justification] — Nine datasets, four IE tasks: LLMs "consistently exhibit inferior performance, higher latency, and increased budget requirements compared to fine-tuned SLMs under most settings". Proposes filter-then-rerank (small model filters, LLM handles only hard samples) with "2.4% F1-gain on average" at "acceptable time and cost" – conceptually close to routing only hard documents to the stronger model.
- ★ **Lamott, M., Weweler, Y.-N., Ulges, A., et al. (2024). LAPDoc: Layout-Aware Prompting for Documents.** ICDAR 2024, LNCS, pp. 142–159. https://doi.org/10.1007/978-3-031-70546-5_9 (arXiv 2402.09841) — [A-3/A-4 analysis] [Evaluation metrics] [VLM vs OCR justification] — Text-only LLMs (ChatGPT-3.5, Solar-70B) prompted with OCR text, with/without layout information. Uses a "type aware accuracy" (dates parsed, currency normalised, case-insensitive strings). Table 2, ChatGPT with plain text: SROIE 79.9, ITInvoices (104 German invoices) 54.5, vs 91.7 / 90.1 for a proprietary KIE model; abstract: layout enrichment helps "by up to 15% compared to just using plain document text".
- **He, J., Wang, L., Hu, Y., et al. (2023). ICL-D3IE: In-Context Learning with Diverse Demonstrations Updating for Document Information Extraction.** ICCV 2023, pp. 19485–19494. https://arxiv.org/abs/2303.05063 — [A-3/A-4 analysis] — GPT-3 (text-davinci-003)/ChatGPT in-context learning on OCR text + boxes, entity-level F1: "ICL-D3IE with GPT-3 (97.88%) outperforms LayoutLMv3base (96.89%) on SROIE"; claims "greater robustness to OCR errors" on perturbed test sets.
- **Gómez, J., Sánchez, J. (2026). Information Extraction from Electricity Invoices with General-Purpose Large Language Models.** arXiv preprint arXiv:2604.25927. https://arxiv.org/abs/2604.25927 — [A-3/A-4

analysis] [Evaluation metrics] — Gemini 1.5 Pro and Mistral-small on 2,400 Spanish invoices (IDSEM), field-level P/R/F1 "based on string matching after normalization" (input: PDF converted to Markdown, not OCR). Mistral-small F1: zero-shot 82.92% → best few-shot 96.11% (Table 8); Gemini 78.39% → 97.61% (Table 7); prompt design mattered more than hyperparameters (">19 percentage points").

- ★ **Šimsa, Š., Šulc, M., Uřičář, M., et al. (2023). DocILE Benchmark for Document Information Localization and Extraction.** ICDAR 2023, LNCS, pp. 147–166. https://doi.org/10.1007/978-3-031-41679-8_9 (arXiv 2302.05658) — [Evaluation metrics] [A-3/A-4 analysis] — 6.7k annotated business documents (mostly invoices), 55 field classes; KILE primary metric is Average Precision, line items use micro F1. Table 3 baseline on provided OCR: RoBERTa-base KILE F1 0.664 / AP 0.534; an "NER upper bound" on the given OCR words reaches only F1 0.946 – OCR tokenisation alone caps achievable accuracy.
- **Stanisławek, T., Graliński, F., Wróblewska, A., et al. (2021). Kleister: Key Information Extraction Datasets Involving Long Documents with Complex Layouts.** ICDAR 2021, LNCS, pp. 564–579. https://doi.org/10.1007/978-3-030-86549-8_36 — [Evaluation metrics] — Two KIE datasets (540 NDAs, 2,788 charity financial reports) with OCR-based baselines; "The best model achieved an 81.77% and an 83.57% F1-score on respectively the Kleister NDA and the Kleister Charity datasets."
- **Palm, R. B., Winther, O., Laws, F. (2017). CloudScan – A Configuration-Free Invoice Analysis System Using Recurrent Neural Networks.** ICDAR 2017, pp. 406–413. https://doi.org/10.1109/ICDAR.2017.74 — [A-3/A-4 analysis] — Pre-LLM invoice extraction from OCR text for "8 important fields using a dataset of 326,471 invoices": "0.891 and 0.887 average F1 scores respectively on seen invoice layouts … unseen invoice layouts … 0.840 average F1 compared to 0.788". Historical baseline for invoice-field F1.
- ★ **Levy, O., Seo, M., Choi, E., Zettlemoyer, L. (2017). Zero-Shot Relation Extraction via Reading Comprehension.** CoNLL 2017, pp. 333–342. https://doi.org/10.18653/v1/K17-1034 — [A-3/A-4 analysis] — Founding reference for the "one question per field" design: "relation extraction can be reduced to answering simple reading comprehension questions, by associating one or more natural-language questions with each relation slot"; zero-shot is "possible, at lower accuracy levels".
- **Wei, X., Cui, X., Cheng, N., et al. (2023). ChatIE: Zero-Shot Information Extraction via Chatting with ChatGPT.** arXiv preprint arXiv:2302.10205. https://arxiv.org/abs/2302.10205 — [A-3/A-4 analysis] — Multi-turn QA formulation of zero-shot IE with ChatGPT; Sec. 4: "ChatIE generally improves performance over six widely used IE datasets by 16.65% points significantly on average" vs single-turn QA.
- **Wang, S., Sun, X., Li, X., et al. (2025). GPT-NER: Named Entity Recognition via Large Language Models.** Findings of NAACL 2025, pp. 4257–4275. https://doi.org/10.18653/v1/2025.findings-naacl.239 (arXiv 2304.10428) — [A-3/A-4 analysis] — NER as generation with entity markers, kNN demonstrations and a self-verification step against hallucinated entities; "90.91 v.s. 92.8 on the full CoNLL2003 dataset" (GPT-3 vs supervised BERT).

4.1 Evaluation metrics

- ★ **Wang, Z., Zhou, Y., Wei, W., et al. (2023). VRDU: A Benchmark for Visually-rich Document Understanding.** KDD 2023, pp. 5184–5193. https://doi.org/10.1145/3580305.3599929 (arXiv 2211.15421) — [Evaluation metrics] — Sec. 4.4 proposes "a type-aware fuzzy matching algorithm" with micro/macro F1, because strict matching is unreasonable: "'$ 40,000' does not match with '40,000' … and 'July 1, 2022' does not match with '07/01/2022'"; prices are converted to numbers and dates parsed before comparison. Maps directly onto the thesis's date and total normalisation.
- ★ **Biten, A. F., Tito, R., Mafla, A., et al. (2019). Scene Text Visual Question Answering.** ICCV 2019. https://arxiv.org/abs/1905.13648 — [Evaluation metrics] — Sec. 3.4 introduces ANLS (1 − normalised Levenshtein distance, set to 0 when the distance reaches $\tau = 0.5$): "We define a threshold τ=0.5"; the metric "will softly penalise the OCR mistakes". Original source for the ANLS used by DocVQA.
- **Mathew, M., Bagal, V., Tito, R., et al. (2022). InfographicVQA.** WACV 2022. https://doi.org/10.1109/WACV51458.2022.00264 (arXiv 2104.12756) — [Evaluation metrics] — Appendix C (Eq. C.1) gives the full ANLS formula: $s = 1 - NL(a, o)$ if $NL < \tau$, else 0, with $\tau = 0.5$ on lower-cased strings; the threshold separates OCR-induced errors from wrong answers. Humans 0.980 ANLS vs best LayoutLM baseline 0.272 (test); OCR by Amazon Textract.

- **ANLS* (Peer et al. 2024)** – full entry in Section 2.1. For metric design: extends ANLS to dictionaries/lists so a whole extracted record (invoice number, date, total) can be scored in one number; Table 3, SROIE "Simple" prompt: gpt-4.1 0.906, Llama-3.1-405B 0.853.
- **Manning, C. D., Raghavan, P., Schütze, H. (2008). Introduction to Information Retrieval, Sec. 8.3 "Evaluation of unranked retrieval sets".** Cambridge University Press (official online edition). https://nlp.stanford.edu/IR-book/html/htmledition/evaluation-of-unranked-retrieval-sets-1.html — [Evaluation metrics] — Textbook definitions: precision "is the fraction of retrieved documents that are relevant", recall "is the fraction of relevant documents that are retrieved", F measure "the weighted harmonic mean of precision and recall"; balanced F1 = 2PR/(P+R) (Eq. 41).

5. Serving: batching, caching, quantization, concurrency

5.1 Batching

- ★ **Crankshaw, D., Wang, X., Zhou, G., et al. (2017). Clipper: A Low-Latency Online Prediction Serving System.** 14th USENIX Symposium on Networked Systems Design and Implementation (NSDI '17), pp. 613–627. https://www.usenix.org/conference/nsdi17/technical-sessions/presentation/crankshaw — [Async/batching] [Caching] [A-3/A-4 analysis] — A serving layer between applications and ML frameworks that uses adaptive (AIMD) batching and a prediction cache. Sec. 4.3: batching "improves throughput and utilization of often costly physical resources such as GPUs, but it does so at the expense of increased latency by requiring all queries in the batch to complete before returning a single prediction"; batching "can improve throughput by up to 26x (the Scikit-Learn SVM in Figure 4) while meeting a 20ms latency SLO". Sec. 4.2: the prediction cache is a function cache keyed on (model, input) with LRU/CLOCK eviction; it "increased feedback processing throughput in Clipper by 1.6x from roughly 6K to 11K observations per second".
- **Olston, C., Fiedel, N., Gorovoy, K., et al. (2017). TensorFlow-Serving: Flexible, High-Performance ML Serving.** NIPS 2017 Workshop on ML Systems; arXiv preprint arXiv:1712.06139. https://arxiv.org/abs/1712.06139 — [Async/batching] — Describes Google's production serving library. Sec. 2.2.1 "Inter-Request Batching" merges requests by concatenating input tensors: "This strategy can boost throughput substantially, but it has to be managed carefully to avoid unduly hurting latency"; per-model batching queues are scheduled "in a round-robin fashion onto a single shared device e.g. GPU" (comparable to Architecture B's single GPU thread).
- **Gujarati, A., Karimi, R., Alzayat, S., et al. (2020). Serving DNNs like Clockwork: Performance Predictability from the Bottom Up.** 14th USENIX Symposium on Operating Systems Design and Implementation (OSDI '20), pp. 443–462. https://www.usenix.org/conference/osdi20/presentation/gujarati — [Async/batching] [A-3/A-4 analysis] — Builds a serving system on the observation that DNN inference has predictable execution time; abstract reports "meeting 100 ms latency targets for 99.9999% of requests". Sec. 5.3: the scheduler "prioritizes larger batch sizes for efficiency", but "Higher batch sizes take longer to execute, so a request close to its deadline might only be satisfiable using a small batch size" (batch sizes 1, 2, 4, 8, 16).
- ★ **NVIDIA (n.d.). Batchers – Triton Inference Server User Guide.** Official documentation, accessed 2026-10. https://docs.nvidia.com/deeplearning/triton-inference-server/user-guide/docs/user_guide/batcher.html — [Async/batching] [A-3/A-4 analysis] — "Dynamic batching is a feature of Triton that allows inference requests to be combined by the server"; "Creating a batch of requests typically results in increased throughput." With `max_queue_delay_microseconds`, "If the delay expires the dynamic batcher sends the batch as is, even though it is not a maximum or preferred size"; tuning advice is to "trade off increased latency for increased throughput" by "increasing delay values until the latency budget is exceeded".
- **Agrawal, A., Kedia, N., Panwar, A., et al. (2024). Taming Throughput-Latency Tradeoff in LLM Inference with Sarathi-Serve.** 18th USENIX OSDI '24, pp. 117–134. https://www.usenix.org/conference/osdi24/presentation/agrawal — [Async/batching] [A-3/A-4 analysis] — Explains that prefill saturates GPU compute while decode has low utilisation, and "batching multiple requests leads to an interleaving of prefill and decode iterations" that makes high throughput and low latency hard to achieve together. Chunked prefills + stall-free scheduling give "2.6x higher serving capacity" for Mistral-7B

on a single A100 and "up to 3.7x" for Yi-34B on two A100s versus vLLM.

- **Daniel, C., Shen, C., Liang, E., Liaw, R. (2023). How continuous batching enables 23x throughput in LLM inference while reducing p50 latency.** Anyscale blog (non-peer-reviewed), 22 June 2023. https://www.anyscale.com/blog/continuous-batching-llm-inference — [Async/batching] — States "LLM inference is memory-IO bound, not compute bound" and that throughput "is largely determined by how large a batch you can fit into high-bandwidth GPU memory"; with static batching the GPU "will be underutilized until the last sequence in the batch finishes generation". Reports "Up to 23x throughput improvement using continuous batching" (OPT-13B, one A100, with vLLM memory optimisations). Use only as an accessible explainer next to Orca/vLLM.

5.2 Routing and cascades (cheap model first)

- **Chen, L., Zaharia, M., Zou, J. (2023). FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance.** arXiv preprint arXiv:2305.05176 (later cited as Transactions on Machine Learning Research 2024, OpenReview id cSimKw5p6R; the OpenReview page could not be opened). https://arxiv.org/abs/2305.05176 — [Compression] [A-3/A-4 analysis] — Proposes prompt adaptation, LLM approximation and an LLM cascade that sends a query to cheaper models first and escalates only when needed. Abstract: FrugalGPT can match GPT-4 "with up to 98% cost reduction" or improve accuracy over GPT-4 by 4% at the same cost. Conceptual precedent for Architecture B's confidence-based routing between a cheap and an accurate model.
- **Ong, I., Almahairi, A., Wu, V., et al. (2025). RouteLLM: Learning to Route LLMs from Preference Data.** International Conference on Learning Representations (ICLR 2025). https://proceedings.iclr.cc/paper_files/paper/2025/hash/5503a7c69d48a2f86fc00b3dc09de686-Abstract-Conference.html — [A-3/A-4 analysis] — Trains routers that choose between a strong and a weak LLM per query. Abstract: routing "can reduce costs by over 2 times without sacrificing response quality". Learned-router counterpart to the hand-set OCR-token/OCR-confidence thresholds in Architecture B.

5.3 Caching

- ★ **Bang, F. (2023). GPTCache: An Open-Source Semantic Cache for LLM Applications Enabling Faster Answers and Cost Savings.** Proceedings of the 3rd Workshop for NLP Open Source Software (NLP-OSS 2023), ACL, pp. 212–218. https://doi.org/10.18653/v1/2023.nlposs-1.24 — [Caching] — Open-source semantic cache that stores LLM responses and serves similar queries without calling the model. Abstract: integrating GPTCache with OpenAI's GPT service "can increase response speed 2-10 times when the cache is hit." Useful contrast: GPTCache keys on embedding similarity, the thesis cache keys on exact image bytes (SHA-256), which cannot return a wrong document's answer.
- **Gill, W., Elidrisi, M., Kalapatapu, P., et al. (2025). MeanCache: User-Centric Semantic Caching for LLM Web Services.** IEEE International Parallel and Distributed Processing Symposium (IPDPS 2025); arXiv:2403.02694. https://arxiv.org/abs/2403.02694 — [Caching] — Client-side semantic cache trained with federated learning; motivates caching because repeated queries "constitute about 31% of the total queries". Reports "approximately 17% higher F-score and a 20% increase in precision" than the prior semantic-cache baseline and "reduces the storage requirement by 83%". Shows that semantic caches have false-hit rates that must be measured (precision), unlike exact-hash caches.
- **National Institute of Standards and Technology (2015). FIPS PUB 180-4: Secure Hash Standard (SHS).** NIST. https://doi.org/10.6028/NIST.FIPS.180-4 (page: https://csrc.nist.gov/pubs/fips/180-4/upd1/final) — [Caching] — Normative definition of SHA-256. Abstract: "This standard specifies hash algorithms that can be used to generate digests of messages" and "The digests are used to detect whether messages have been changed since the digests were generated." Justifies content-addressed cache keys.
- **Python Software Foundation (n.d.). hashlib – Secure hashes and message digests.** Python 3 official documentation. https://docs.python.org/3/library/hashlib.html — [Caching] — Provides "the FIPS secure hash algorithms SHA224, SHA256, SHA384, SHA512, (defined in the FIPS 180-4 standard)"; `hashlib.file_digest()` (3.11+) hashes a file object directly. Warns that MD5/SHA1 have known collision weaknesses, supporting the choice of SHA-256.

- **SQLite (n.d.). Write-Ahead Logging.** Official documentation. https://www.sqlite.org/wal.html — [Caching] — Sec. 1: "WAL is significantly faster in most scenarios" and "WAL provides more concurrency as readers do not block writers and a writer does not block readers." Sec. 2.2: "there can only be one writer at a time" – relevant if several OCR worker processes write cache rows.
- **SQLite (n.d.). SQLite As An Application File Format.** Official documentation. https://www.sqlite.org/appfileformat.html — [Caching] — "An SQLite database is contained in a single file, which is easily copied or moved or attached"; transactions "either happen completely or not at all, even during system crashes or power failures." Justifies SQLite as a durable single-file result cache.

5.4 Quantization

- ★ **Hugging Face (n.d.). Optimizing LLMs for Speed and Memory.** Hugging Face Transformers documentation. https://huggingface.co/docs/transformers/llm_tutorial_optimization — [Compression] [A-3/A-4 analysis] [VLM vs OCR justification] — Rule of thumb: "Loading the weights of a model having X billion parameters requires roughly 4 * X GB of VRAM in float32 precision" (2 * X GB in bf16/fp16); for inputs "less than 1024 tokens" memory is "very much dominated by the memory requirement to load the weights". On quantization: computation is `Y = X * dequantize(W)`, "Therefore, inference time is often **not** reduced when using quantized weights, but rather increases"; octocoder (~15B) went from ~31 GB to 15.2 GB (8-bit, "slight slow-down") and 9.5 GB (4-bit, "again a bit slower"). Conclusion: "model quantization trades improved memory efficiency against accuracy and in some cases inference time." Directly matches the thesis result (memory saved, no time saved).
- ★ **Belkada, Y., Sun, M., Moutawwakil, I., et al. (2023). Overview of natively supported quantization schemes in Transformers.** Hugging Face blog (non-peer-reviewed), 12 Sep 2023. https://huggingface.co/blog/overview-quantization-transformers — [Compression] [A-3/A-4 analysis] — Table "Inference speed (forward pass only)" (Llama-2-13b, A100-80GB, prompt length 512): batch 1 fp16 36.958 ms/token vs bitsandbytes 4-bit 52.00 ms/token (peak memory 29152.98 MB vs 11018.36 MB); batch 16 fp16 69.94 vs bitsandbytes 113.98 ms. States "bitsandbytes 4-bit models are slow compared to GPTQ when using generate."
- **Belkada, Y., Dettmers, T. (2022). A Gentle Introduction to 8-bit Matrix Multiplication for transformers at scale using Hugging Face Transformers, Accelerate and bitsandbytes.** Hugging Face blog (non-peer-reviewed), 17 Aug 2022. https://huggingface.co/blog/hf-bitsandbytes-integration — [Compression] [A-3/A-4 analysis] — Section "Is it faster than native models?": "BLOOM-176B with LLM.int8() is about 15% to 23% slower than the fp16 version" and "larger slowdowns for smaller models, like T5-3B and T5-11B". Table: T5-3B batch 1, 45 ms/token fp32 (2x T4) vs 312 ms/token int8 (1x T4); batch 8: 7.2 vs 39.1 ms. Caveat: different GPU counts per row, and the text gives 173 ms for T5-3B int8, which does not match the table. Closest published analogue (T5 family, fp32 baseline) to the thesis result.
- **Belkada, Y., Dettmers, T., Pagnoni, A., et al. (2023). Making LLMs even more accessible with bitsandbytes, 4-bit quantization and QLoRA.** Hugging Face blog (non-peer-reviewed), 24 May 2023. https://huggingface.co/blog/4bit-transformers-bitsandbytes — [Compression] [A-3/A-4 analysis] — Mechanism: "While 4-bit bitsandbytes stores weights in 4-bits, the computation still happens in 16 or 32-bit"; default compute dtype is "torch.float32", and computation "will be faster if one uses a 16-bit compute dtype". Explains why NF4 with fp32 compute saves memory but not time.
- **Hugging Face (n.d.). Bitsandbytes – Transformers quantization documentation.** Official documentation. https://huggingface.co/docs/transformers/main/en/quantization/bitsandbytes — [Compression] — "Quantizing a model in 4-bit reduces your memory-usage by 4x"; recommends changing compute dtype "from float32 (the default value) to bf16 in BitsAndBytesConfig to speedup computation"; NF4 is "adapted for weights initialized from a normal distribution".
- ★ **Dettmers, T. (2023). bitsandbytes 0.40.0 release notes.** GitHub release, official project page. https://github.com/bitsandbytes-foundation/bitsandbytes/releases/tag/0.40.0 — [Compression] [A-3/A-4 analysis] — "Added 4-bit inference kernels for batch size=1. Currently supported are the NF4, FP4 data types." Expected speedups vs 16-bit are stated only "for matrix multiplications with inner product dimension of at least 4096" ("2.2x for Turing", "3.4x for Ampere", "4.0x for Ada/Hopper"). Flan-T5-base (d_model 768, d_ff 2048 per its config.json, https://huggingface.co/google/flan-t5-base/blob/main/config.json) run at batch 8 meets neither condition – consistent with the absence of a speed-up in the thesis.

- **Dettmers, T., Zettlemoyer, L. (2023). The case for 4-bit precision: k-bit Inference Scaling Laws.** ICML 2023, PMLR 202:7750–7774. https://proceedings.mlr.press/v202/dettmers23a.html — [Compression] [A-3/A-4 analysis] — "4-bit precision is almost universally optimal for total model bits and zero-shot accuracy." Sec. 2.1: reduction in total model bits "is strongly correlated with inference latency for small inference batch sizes"; the limitations note that "low-bit models with 16-bit inputs might be less latency efficient" for large mini-batches.
- ★ **Gholami, A., Kim, S., Dong, Z., et al. (2021). A Survey of Quantization Methods for Efficient Neural Network Inference.** arXiv preprint arXiv:2103.13630 (also a chapter in *Low-Power Computer Vision*, CRC Press 2022). https://arxiv.org/abs/2103.13630 — [Compression] [A-3/A-4 analysis] — Sec. IV-A defines simulated (fake) quantization: parameters "are stored in low-precision, but the operations … are carried out with floating point arithmetic", so they "need to be dequantized before the floating point operations"; "As such, one cannot fully benefit from fast and efficient low-precision logic with simulated quantization." Clearest peer-level explanation of dequantization overhead.
- **Lin, J., Tang, J., Tang, H., et al. (2024). AWQ: Activation-aware Weight Quantization for On-Device LLM Compression and Acceleration.** Proceedings of Machine Learning and Systems 6 (MLSys 2024; Best Paper Award per the official repository). https://proceedings.mlsys.org/paper_files/paper/2024/hash/42a452cbafa9dd64e9ba4aa95cc1ef21-Abstract-Conference.html — [Compression] — Weight-only low-bit quantization based on the observation that "protecting 1% of salient weights can greatly reduce quantization error". The TinyChat kernels give "more than 3x speedup over the Huggingface FP16 implementation" – i.e. the speed-up comes from custom fused kernels, not from 4-bit storage alone.
- **Xiao, G., Lin, J., Seznec, M., et al. (2023). SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models.** ICML 2023, PMLR 202:38087–38099. https://proceedings.mlr.press/v202/xiao23c.html — [Compression] — Training-free W8A8 quantization that moves quantization difficulty from activations to weights; abstract reports up to 1.56x speed-up and 2x memory reduction with "negligible loss in accuracy". Speed-up requires integer arithmetic for both weights and activations, unlike bitsandbytes weight-only NF4.
- **Hugging Face (n.d.). GGUF – Hub documentation;** and **ggml-org (n.d.). llama.cpp README.** Official documentation. https://huggingface.co/docs/hub/gguf ; https://github.com/ggml-org/llama.cpp — [Compression] — GGUF is "a binary format that is optimized for quick loading and saving of models"; Q4_K is a super-block 4-bit format "resulting in 4.5 bits-per-weight". llama.cpp offers "1.5-bit, 2-bit, 3-bit, 4-bit, 5-bit, 6-bit, and 8-bit integer quantization for faster inference and reduced memory use". Relevant as the consumer-hardware alternative (CPU/GPU) for running quantized decoder LLMs/VLMs.

5.5 Python concurrency

- ★ **Python Software Foundation (n.d.). Event Loop – asyncio:** `loop.run_in_executor`**.** Python 3 official documentation. https://docs.python.org/3/library/asyncio-eventloop.html — [Async/batching] — `loop.run_in_executor(executor, func, *args)` will "Arrange for func to be called in the specified executor" and returns an awaitable future. The official example states: "CPU-bound operations will block the event loop: in general it is preferable to run them in a process pool."
- **Python Software Foundation (n.d.). concurrent.futures – Launching parallel tasks.** Python 3 official documentation. https://docs.python.org/3/library/concurrent.futures.html — [Async/batching] — "ProcessPoolExecutor uses the multiprocessing module, which allows it to side-step the Global Interpreter Lock but also means that only picklable objects can be executed and returned." Default `max_workers` is `os.process_cpu_count()`.
- **Python Software Foundation (n.d.). Glossary: global interpreter lock.** Python 3 official documentation. https://docs.python.org/3/glossary.html — [Async/batching] — GIL: "The mechanism used by the CPython interpreter to assure that only one thread executes Python bytecode at a time", which comes "at the expense of much of the parallelism afforded by multi-processor machines."
- **Gross, S. (2023). PEP 703 – Making the Global Interpreter Lock Optional in CPython.** Python Enhancement Proposal (Final; Python 3.13). https://peps.python.org/pep-0703/ — [Async/batching] — Abstract: "CPython's global interpreter lock ("GIL") prevents multiple threads from executing Python code at the same time" and "The GIL is an obstacle to using multi-core CPUs from Python efficiently." The Motivation

quotes a user blocked "by the prohibitive IPC costs" of subprocesses – supports discussing pickling/IPC overhead of the OCR process pool.

6. Latency and throughput theory

- ★ **Little, J. D. C. (2011). Little's Law as Viewed on Its 50th Anniversary.** Operations Research 59(3), 536–549. https://doi.org/10.1287/opre.1110.0940 (text read from a course-hosted copy: https://people.cs.umass.edu/~emery/classes/cmpsci691st/readings/OS/Littles-Law-50-Years-Later.pdf) — [A-3/A-4 analysis] — Sec. 1: "the average number of items in a queuing system, denoted L, equals the average arrival rate of items to the system, λ, multiplied by the average waiting time". Sec. 4.1 maps it to computer systems: L = "average number of requests in process", λ = "average arrival rate of requests (requests/second)", W = "average response time per request (seconds) = latency". Sec. 2: over a finite observation window the relation is "numerically exact" – applicable to a benchmark run of 600 documents.
- ★ **Amdahl, G. M. (1967). Validity of the single processor approach to achieving large scale computing capabilities.** AFIPS Spring Joint Computer Conference, vol. 30, pp. 483–485. https://doi.org/10.1145/1465482.1465560 (text: https://www3.cs.stonybrook.edu/~rezaul/Spring-2019/CSE613/reading/Amdahl-1967.pdf) — [A-3/A-4 analysis] — Sequential "data management housekeeping ... accounts for 40% of the executed instructions", and "Overhead alone would then place an upper limit on throughput of five to seven times the sequential processing rate". Basis for explaining why adding OCR workers is capped by the serial GPU stage.
- **Williams, S., Waterman, A., Patterson, D. (2009). Roofline: an insightful visual performance model for multicore architectures.** Communications of the ACM 52(4), 65–76. https://doi.org/10.1145/1498765.1498785 (text: tech report UCB/EECS-2008-134, https://www2.eecs.berkeley.edu/Pubs/TechRpts/2008/EECS-2008-134.html) — [A-3/A-4 analysis] — "Attainable GFlops/sec = Min(Peak Floating Point Performance, Peak Memory Bandwidth x Operational Intensity)"; kernels with low operational intensity are "ultimately memory bound". Explains why batching (more work per weight load) raises GPU throughput.
- **Yuan, Z., Shang, Y., Zhou, Y., et al. (2024). LLM Inference Unveiled: Survey and Roofline Model Insights.** arXiv preprint arXiv:2402.16363. https://arxiv.org/abs/2402.16363 — [Compression] [A-3/A-4 analysis] — Sec. 2.2: "during the prefill stage, the majority of computations are compute-bound ... in the decode stage, all computations are memory-bound". Sec. 3.1.1: quantization helps memory-bound layers, but when operational intensity "remains within the compute-bound range" "there is no performance improvement"; recommends "quantization, kernel fusion and increasing batch size" for memory-bound layers.
- **Dean, J., Barroso, L. A. (2013). The Tail at Scale.** Communications of the ACM 56(2), 74–80. https://research.google/pubs/the-tail-at-scale/ — [A-3/A-4 analysis] — "It is challenging to keep the tail of the latency distribution low for interactive services"; proposes "latency tail-tolerant" techniques that "create a predictably responsive whole out of less predictable parts". Supports reporting p95/p99 latency, not just means.
- **Schwartz, R., Dodge, J., Smith, N. A., Etzioni, O. (2020). Green AI.** Communications of the ACM 63(12), 54–63. https://doi.org/10.1145/3381831 (text: arXiv:1907.10597) — [A-3/A-4 analysis] [VLM vs OCR justification] — Argues for "making efficiency an evaluation criterion for research alongside accuracy and related measures" and for reporting the computational "price tag"; cites an estimated 300,000x increase in deep-learning compute from 2012 to 2018. Justifies reporting throughput/latency/memory next to F1. (CACM page returned 403; metadata via Crossref, content via arXiv.)

7. Sources that could not be verified

UNVERIFIED (could not be opened, or only bibliographic metadata was available):

- **Little, J. D. C. (1961). A Proof for the Queuing Formula: L = λW.** Operations Research 9(3), 383–387. https://doi.org/10.1287/opre.9.3.383 — Metadata confirmed via Crossref only; INFORMS page returned 403, full text not read. Cite Little (2011) (Section 6) for the wording; add 1961 only as the original-proof reference.

- **Levenshtein, V. I. (1966). Binary codes capable of correcting deletions, insertions, and reversals.** Soviet Physics Doklady 10(8), 707–710. — Original not opened; bibliographic entry and definition confirmed at NIST DADS (https://xlinux.nist.gov/dads/HTML/Levenshtein.html: "The smallest number of insertions, deletions, and substitutions required to change one string or tree into another"). Standard citation for edit distance; safe to cite but not read.
- **Hamdi, A., Linhares Pontes, E., Sidere, N., Coustaty, M., Doucet, A. (2023). In-depth analysis of the impact of OCR errors on named entity recognition and linking.** Natural Language Engineering 29(2), 425–448. https://doi.org/10.1017/S1351324922000110 — Paywalled; title, authors and DOI confirmed via Crossref; no content read.
- **Zhong, Y., et al. (2024). DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving.** USENIX OSDI '24, pp. 193–210. — USENIX page opened, but the reported numbers came only from a summary and were not checked verbatim; not used.
- **Sokolova, M., Lapalme, G. (2009). A systematic analysis of performance measures for classification tasks.** Information Processing & Management 45(4), 427–437. https://doi.org/10.1016/j.ipm.2009.03.002 — Metadata via Crossref only; ScienceDirect returned 403. Use Manning et al. 2008 (Section 4.1) instead for F1 definitions.
- **Cuper, M., van Dongen, C., Koster, T. (2023). Unraveling Confidence: Examining Confidence Scores as Proxy for OCR Quality.** ICDAR 2023, LNCS, pp. 104–120. https://doi.org/10.1007/978-3-031-41734-4_7 — Metadata confirmed via Crossref; found via search (reported 83% accuracy separating very good from very bad OCR, calibration per engine), but neither the KB lab page (403) nor the paper was opened. UNVERIFIED.
- **Meluš, D. (2024). Enhancing Quality of Optical Character Recognition for Financial Document Processing.** Master's thesis, Masaryk University (is.muni.cz/th/o88qt) — repository behind a JavaScript bot challenge. UNVERIFIED.
- **Lehtonen, R. (2020). Automated classification of receipts and invoices along with document extraction.** Master's thesis, University of Turku (utupub) — repository returned 403. UNVERIFIED.
- **Ferreira de Almeida, B. (2026). AI Invoice Validator: Hybrid ML and LLM-Based Electricity Invoice Information Extraction.** Master's thesis, University of Porto (FEUP) — title/author/date confirmed, but embargoed until 2027 and no abstract. PARTIAL.
- **Seen in the DiVA/Uppsök index, abstracts not read:** Fredriksson (KTH 2023, invoice classification with LayoutLMv3); Kadir (Uppsala 2022, invoice line items with SaaS models such as Textract); Dwyer & Bertse (KTH 2022, ML for OCR post-processing of invoices); Andersson (Uppsala 2020, invoice extraction as a constraint-satisfaction problem).
- **Not found:** a paper titled "Is GPT-4V a good document understanding model"; any peer-reviewed Tesseract+LLM vs VLM comparison on invoices/receipts with measured latency on consumer hardware (this gap is itself a point the thesis can make).

Verified, but with a caveat on venue or a specific number:

- FrugalGPT – arXiv verified; the TMLR 2024 publication is taken from later citations because the OpenReview page could not be opened.
- AWQ "Best Paper Award at MLSys 2024" – stated only in the official GitHub README, not checked on an MLSys awards page.
- Hemmer et al. 2024 – DAS 2024 acceptance seen only on the IAPR DAS site; cite as arXiv.
- Green AI (CACM) and The Tail at Scale (CACM), Roofline (CACM), Amdahl (AFIPS) – publisher pages returned 403; content read from arXiv / official Google Research page / UC Berkeley tech report / a reprint, metadata from Crossref.
- Florence-2 – TextVQA is 81.5 in the text but 73.5 in Table 5; check the PDF before quoting.
- MiniCPM-V "640 tokens for a 1.8M-pixel image" – from the MiniCPM-V 2.6 model card, not the arXiv paper.
- PaliGemma 2 CPU timings (Table 9) were garbled in HTML extraction; only the 12.1 GB / 4.0 GB memory figures are reliable from this check.
- Belkada & Dettmers 2022 (HF 8-bit blog) – table rows use different GPU counts (fp32 on 2x T4, int8 on 1x T4), and the text's 173 ms for T5-3B int8 differs from the table's 312 ms.

- "The Tail at Scale" example "1-in-100 slow requests → 63%" – not visible on any page opened; not used.
- Opened but excluded as too weak: Gaikwad (2026, arXiv 2604.25176; ground truth from OCR-ensemble voting), Yashwant et al. (2025, arXiv 2510.15727) and Khanchandani et al. (2025, arXiv 2511.05547) – invoice LLM pipelines without numbers in the abstract; López Tello (UPV 2025) thesis – no results in the abstract, PDF closed; Zahi (Lund 2026) – Kubernetes LLM inference, mentions caching only as a feature.